

Potential Effects of Sea Level Rise and High Tide Flooding on *Laterallus jamaicensis jamaicensis* (Eastern Black Rail) Coastal Breeding Areas

Chapter F of
**Effects of Climate Change on Fish and Wildlife Species in
the United States**

Open-File Report 2021–1104–F

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By Catherine A. Nikiel and Marta P. Lyons

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**U.S. Department of the Interior
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Conversion Factors

U.S. customary units to International System of Units

Multiply	By	To obtain
Length		
foot (ft)	0.3048	meter (m)

International System of Units to U.S. customary units

Multiply	By	To obtain
Length		
centimeter (cm)	0.3937	inch (in.)
meter (m)	3.281	foot (ft)
meter (m)	1.094	yard (yd)
Flow rate		
millimeter per year (mm/yr)	0.03937	inch per year (in/yr)

Datums

Vertical coordinate information is referenced to the North American Vertical Datum of 1988 (NAVD 88).

Horizontal coordinate information is referenced to the North American Datum of 1983 (NAD83; 2011)/Albers Equal-Area Conic projection (EPSG: 6350).

Elevation, as used in this report, refers to distance above the vertical datum.

Abbreviations

±	plus or minus
C-CAP	Coastal Change Analysis Program
GMSL	global mean sea level
NLCD	National Land Cover Database
NOAA	National Oceanic and Atmospheric Administration
RCP	representative concentration pathway
SLR	sea level rise
SSP	Shared Socioeconomic Pathway

Potential Effects of Sea Level Rise and High Tide Flooding on *Laterallus jamaicensis jamaicensis* (Eastern Black Rail) Coastal Breeding Areas

By Catherine A. Nikiel^{1,2} and Marta P. Lyons¹

Abstract

Laterallus jamaicensis jamaicensis (eastern black rails; Gmelin, 1789) are facing increasing risk from flooding in coastal breeding habitats because of rising sea levels combined with standard high tide flooding. In this report, we examine regional differences in relative rates of sea level rise, days in the breeding season above historical high tide flooding thresholds, future inundation of current (2021) emergent wetlands, and potential marsh resiliency for the breeding distribution of the eastern black rail across the Atlantic and U.S. Gulf coasts. By midcentury (2050), two sea level rise scenarios (intermediate low and intermediate) indicate that areas analyzed in Texas and the Mid-Atlantic will experience at least minor flood levels for more than half of the breeding season. By the end of the century (2100), all tidal gages in the Atlantic and U.S. Gulf coasts are projected to experience at least moderate flood levels for most of the current (April–September) eastern black rail breeding season. In some areas like New Jersey, this translates to inundation for most of the emergent wetlands in the representative parishes and counties analyzed in this report. In other parts of the coastal distribution, estimates of increases in inundation are lower or more variable, stemming from differences in the elevation of existing emergent marsh, especially at the herbaceous wetland/woody wetland transition zone. Sea level rise and tidal flooding are not projected to pose an equal risk across the coastal distribution of the eastern black rail, leading to variation in risk of nest loss because of flooding. The degree to which these wetlands and birds will adapt to changing sea level and salinity depends on a range of factors including future expansion of developed areas and the ability of marsh areas to move inland. Restoration and active management of coastal wetland areas may be necessary to maintain appropriate breeding habitat.

Purpose and Scope

Laterallus jamaicensis jamaicensis (eastern black rail), a subspecies of *Laterallus jamaicensis* (black rail), were listed as threatened under the Endangered Species Act in 2020. Habitat loss, sea level rise (SLR) and tidal flooding, and storm intensity and frequency were identified as some of the primary threats to the eastern black rail and reasons for the listing decision (U.S. Fish and Wildlife Service, 2020). In subsequent years, extensive research including new distribution modeling has been completed to characterize the environmental needs of these birds now and into the future (Watts and Beisler, 2021); however, the secretiveness of eastern black rails has made assessing some of the current and future threats that are likely to affect this subspecies difficult.

The eastern black rail's preferred nesting habitat is emergent wetlands, where dense vegetation provides protection from predators. In coastal wetlands, nesting occurs in the high marsh where the soil is wet or the water is shallow. This area may be inundated during lunar and wind-driven tidal events but is at a higher elevation than the area that is inundated by daily tides (Watts, 2016; Stevens and others, 2022). Individual studies have identified nest flooding as a substantial source of mortality for eastern black rails (Hand and others, 2021). Future nest flooding risk is projected to increase because of the combined effects of SLR, high tides, and storm events; however, little information is available on the exact location and success of nest sites with respect to elevation across the entire coastal distribution. Without data on nesting outcomes after flooding events to parameterize a model, quantifying current and future flooding effects on nests at a finely resolved spatial scale is not feasible. Additionally, the scale at which nest flooding occurs in a heterogeneous landscape (for example, habitat and elevation) and the coarse estimates of SLR and tidal height make it difficult to quantify exact flooding risk at a nesting-relevant scale. It is still possible to highlight areas that may face climate stressors compounded with existing habitat challenges.

To assess potential changes to coastal nesting habitat, we used ecological studies for the eastern black rail (Hand and others, 2021; Watts and Beisler, 2021; Stevens and

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2 Potential Effects of Sea Level Rise and High Tide Flooding on *Laterallus jamaicensis jamaicensis*

others, 2022) and *Ammospiza caudacuta* (saltmarsh sparrow; Gmelin, 1788), a species with similar habitat requirements (Bayard and Elphick, 2011; Field and others, 2017); research on marsh resiliency and marsh migration; and projections for SLR. Our goal in this report is to provide a qualitative analysis of potential risks because of environmental changes (relative SLR and high tide flood frequency) in the absence of the necessary data (exact nesting locations, information on renesting frequency and success, and local marsh migration projections) for inclusion in a quantitative model.

The eastern black rail is a cryptic marsh bird that, in the United States, primarily inhabits the Atlantic and U.S. Gulf coasts (fig. 1) with some inland populations in

the Great Plains (not shown; Watts, 2016). Coastal parishes and counties that have ocean-connected land less than or equal to 10 feet (ft) (referenced to North American Vertical Datum of 1988) above sea level are denoted on figure 1. For the purposes of this report, which explores changes in tidal flooding caused by climate change, we focus on the coastal distribution of the subspecies, particularly in areas with recent breeding activity (Watts, 2016); six county clusters where gages have relative SLR projections were chosen from these areas to demonstrate a variety of vulnerability (fig. 1).

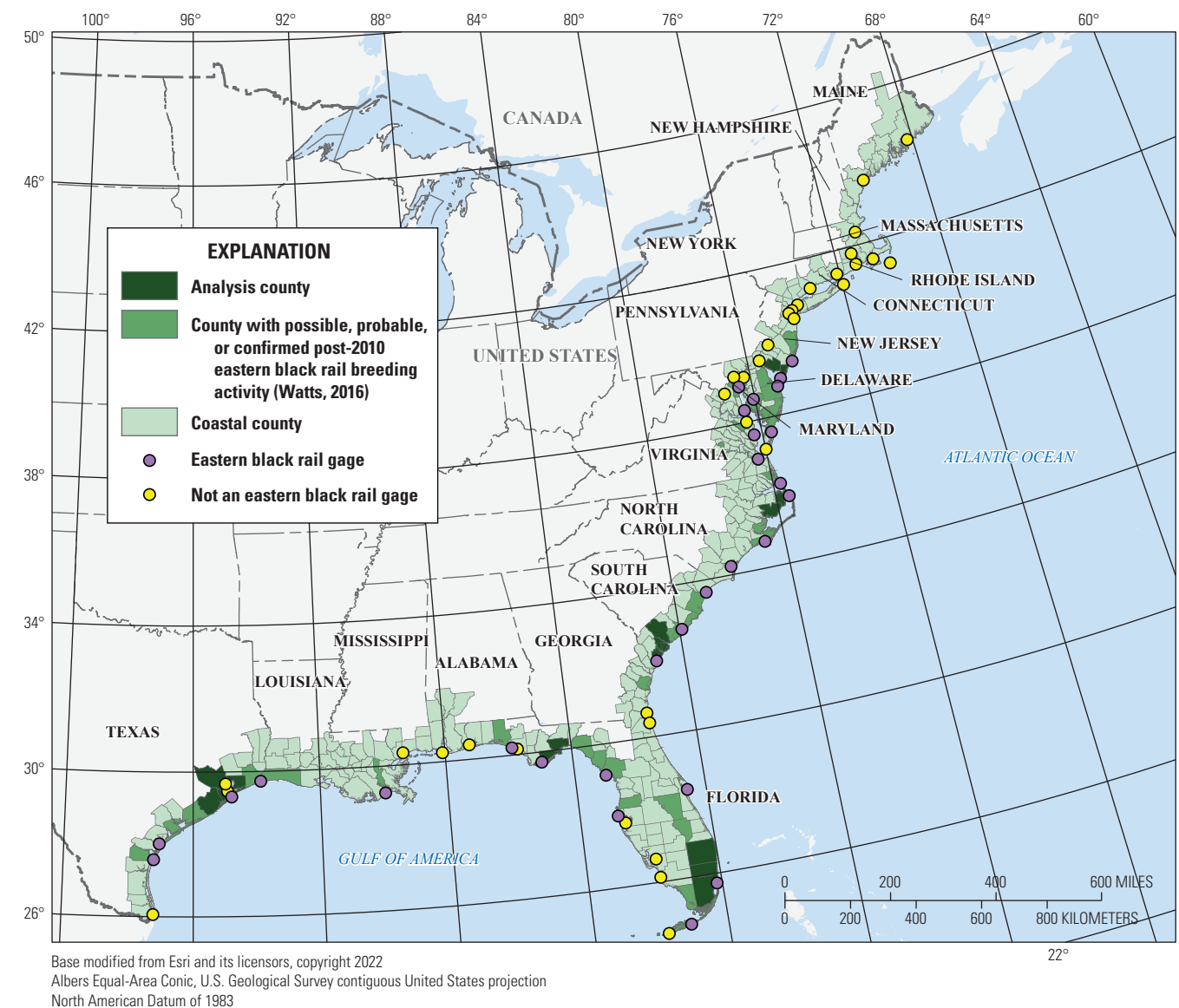


Figure 1. Map showing *Laterallus jamaicensis jamaicensis* (eastern black rail) coastal breeding range.

Eastern Black Rail Ecology

The eastern black rail's secretive nature has made this subspecies difficult to study (Roach and Barrett, 2015). The studies that provide background for this report primarily used vocalizations to identify rail occurrence. These data do not provide information on breeding ecology including population responses to environmental factors like nest flooding (Watts and Beisler, 2021). Recent advances in camera trap data provide more insight into eastern black rail habitat use but have only been deployed in limited parts of the subspecies' range (Hand and others, 2021).

The wide but patchy geographic distribution (Stevens and Conway, 2021) of the eastern black rail makes it difficult to comprehensively evaluate subspecies status with relation to their current and future environment; however, surveys and range-wide studies have broadly identified the conditions necessary for eastern black rails (see Watts and Beisler [2021] for recent advances). The eastern black rail is a wetland-dependent bird, and occupancy models indicate that this subspecies prefers high marsh, occupying the transition zone at the edge of nonforested wetlands where dense vegetation provides protection from predators (Stevens and others, 2022) and where elevation provides protection from tidal inundation except during extreme tides (Watts, 2016). Eastern black rails nest and use areas with shallow water or wet soils that stay wet even during dry periods (Hines and others, 2024) and locations where they can build nests high enough to prevent flooding, avoiding low tidal marshes (Watts, 2016; Haverland and others, 2021). Several studies on eastern black rail locations note the importance of this high marsh (Haverland and others, 2021; Stevens and others, 2022).

Flooding Risk to Eastern Black Rail

Because of their reliance on coastal marsh habitat, eastern black rails are at risk from tidal flooding. Eastern black rails are most vulnerable to flooding during nest incubation, brooding, and molting periods. Across their distribution, egg laying and incubation occur between March and August; nesting peaks around the third week of June with no significant difference in timing by region (Watts, 2021). Although documentation on nest outcomes is limited, where it has been studied, nest flooding was the greatest cause of egg loss for the eastern black rail (Legare and Eddleman, 2001). The combined effects of rising sea levels with high tides, particularly during nesting season, have been identified as some of the greatest threats to the subspecies' productivity and survival (Hand and others, 2021; Watts and others, 2021).

The risk and response of the eastern black rail to nest flooding and increasing flood potential have not been extensively studied. Another coastal wetland bird species with overlapping distribution that also nests in the high marsh, the saltmarsh sparrow, has received more attention. Nest flooding

is a major source of nest failure for the saltmarsh sparrow; inundation depth and frequency of nest flooding are strongly correlated with nest failure (Bayard and Elphick, 2011). Population models that incorporate SLR indicate a reduction in successful reproduction to the point of extinction in the absence of marsh migration (Field and others, 2017). For Maryland, Audubon Maryland-DC developed a tool to identify areas for marsh conservation and restoration in the face of SLR specifically targeting the high marsh system used by saltmarsh sparrows (Lerner and others, 2013). The future changes to breeding habitat identified for the saltmarsh sparrow are likely mirrored in future risks to the eastern black rail.

Analysis Range and Habitat Selection

To investigate the future risk of flooding in eastern black rail breeding areas, we determine a range of parishes and counties along the Atlantic and U.S. Gulf coasts for analysis and further restrict analyses to certain land cover types and seasons. Analysis was limited to April–September (a total of 183 days). This period includes likely timing of egg laying, incubation, hatching, and then adult molting when eastern black rails are at risk from inundation (Hand and others, 2021).

Potential habitat is delineated two ways: (1) a conservative approach considers all areas in a county in the 30-meter (m) resolution National Land Cover Database (NLCD) dataset designated as emergent herbaceous wetland (NLCD class 95) to be potential habitat (Dewitz, 2023) and (2) a second designation considers only the transition zone, which is the part of emergent herbaceous wetland within 500 m of woody wetland (NLCD class 90) as potential habitat. The second approach has a reduced absolute area, but inundation percentages tend to be lower because this transition zone is more inland. We compared the NLCD emergent wetland data layer with wetland layers in other products including the 30-m resolution Coastal Change Analysis Program (C-CAP) 2016 Regional Land Cover dataset and the most recent National Wetlands Inventory vector dataset (Office for Coastal Management, 2024; U.S. Fish and Wildlife Service, 2024). Agreement in the historical distribution of wetland among all three datasets was consistent. The added marsh type delineation in C-CAP and National Wetlands Inventory was ultimately deemed unnecessary given the lack of specific nesting site data, and NLCD emergent herbaceous wetland was the delineation used in this analysis. Several studies on eastern black rail locations note the importance of high marsh (Haverland and others, 2021; Stevens and others, 2022), and multiple recent datasets characterize this classification covering sections of the Atlantic and U.S. Gulf coasts using different methodologies (Allen and others, 2017; Correll and others, 2019; Enwright and others, 2022);

however, this report uses a spatially consistent dataset (the NLCD dataset) to demonstrate the inundation risk across parishes and counties of interest.

For the scope of this report, we focus on coastal parishes and counties with possible, probable, or confirmed eastern black rail breeding activity post-2010, as listed in Watts (2016) and identified in figure 1. Additional parishes and counties with ocean-connected land 10 ft or less in elevation without recent reported eastern black rail activity are identified (U.S. Geological Survey, 2024). Ten feet was chosen because of the range of projected SLR available from the National Oceanic and Atmospheric Administration (NOAA) Sea Level Rise Viewer (National Oceanic and Atmospheric Administration, 2024). Six regions with recent breeding activity were selected to assess how SLR, high tide flooding frequency, and land cover distribution will affect historical habitat inundation across the range (fig. 1).

Relative Sea Level Rise and High Tide Flooding Levels

SLR is driven by changes in climate, and SLR scenarios follow from emissions scenarios; however, there is no direct one-to-one correspondence between a single emissions pathway used to discuss climate projections (for example, Shared Socioeconomic Pathway [SSP] 2–4.5) and a single scenario of SLR. Because of a combination of processes that may occur within several emissions scenarios and the inclusion of lower confidence processes related to ice sheets,

SLR scenarios are assigned a level of probability of occurring under a given emissions scenario (fig. 2). In this report, we considered two SLR scenarios that bound the likely range of SLR for the moderate and high emissions scenarios (representative concentration pathway [RCP] 4.5 and 8.5, respectively) and the median of the 2050 extrapolation-based estimates of global mean sea level (GMSL); these SLR scenarios are the intermediate low (0.5 m of GMSL rise by 2100) and the intermediate scenarios (1 m of GMSL rise by 2100) (Sweet and others, 2018, 2022). The intermediate low SLR scenario has the greatest probability of occurring under an intermediate emissions scenario (for example SSP2–4.5 or RCP 4.5) when considering medium-high (dark yellow in fig. 2) and low (light yellow in fig. 2) confidence processes. Low confidence processes are Antarctic and Greenland ice-sheet processes with low agreement regarding rates, magnitude, and thresholds of change (Sweet and others, 2022). The intermediate SLR scenario has the greatest probability of occurring under a high emissions scenario (for example SSP5–8.5 or RCP 8.5) when low confidence processes (light red in fig. 2) are disregarded. The use of multiple scenarios allows for more comprehensive risk assessment, especially at end of century. The use of other SLR scenarios such as intermediate high (1.5 m of GMSL rise by 2100) and high (2 m of GMSL rise by 2100) will exacerbate risks detailed here because sea levels would rise more rapidly and to a higher point by 2100.

These SLR scenarios represent a single average value of GMSL, which is driven by thermal ocean expansion and contributions from melting ice; however, the true SLR experienced in a location—relative SLR—is driven by

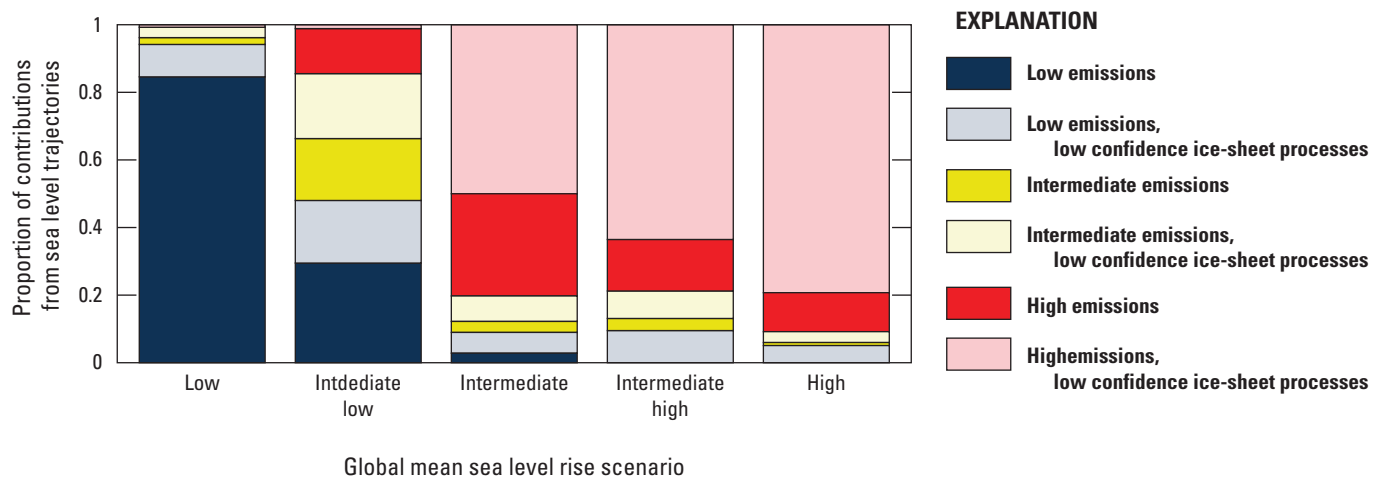


Figure 2. Graph showing proportions of the contributions from different Intergovernmental Panel on Climate Change Annual Report 6 sea level trajectories to each of the five global mean sea level rise scenarios used in Sweet and others (2022). The emissions pathways associated with the sea level trajectories are as follows: low emissions (Shared Socioeconomic Pathway [SSP] 1–1.9 or SSP1–2.6), intermediate emissions (SSP2–4.5), high emissions (SSP3–7.0 or SSP5–8.5). Shifts among different global mean sea level rise scenarios approximately reflect the relative odds of being close to a given scenario under different emissions pathways; for example, the low scenario is much more plausible under a low emissions pathway, whereas the intermediate and high scenarios are much more likely to be associated with high emissions pathways and with low-confidence ice-sheet processes (modified from Sweet and others [2022]).

additional local processes, including subsidence or accretion of coastal land, land water storage, stereodynamic effects from shifting wind and ocean currents, and gravitational and deformation driven changes from ice-mass loss (Sweet and others, 2022). These processes occur across multiple scales and can cause sea level variability from year to year. Data are presented here for the projected relative SLR at 2100 under the intermediate low and intermediate SLR scenarios (figs. 3 and 4; Sweet and others, 2022) relative to the historical Mean Higher High Water in 2000. Figures for relative SLR at 2050 are available in [appendix 1](#); however, users should be aware of the uncertainty in these projected levels ([app. 1](#)) and that a given sea level may occur sooner or later at a given location. Mean Higher High Water is set as the threshold that is

exceeded by water levels about 50 plus or minus ($\pm$) 5 percent of the days per year at a location within the historical tidal epoch (1983–2001).

Relative sea level is calculated at tidal gages, but the density of gages differs across the coastline. The Interagency Task Force on Sea Level Rise developed a representation of relative sea level on a 1-degree grid to provide more even coverage across the coast, and this product is recommended when looking at scenarios in areas without a tidal gage (county scale or greater; Collini and others, 2022). We include relative sea level and high tide flooding days at the tidal gage locations ([fig. 1](#)). This is the scale at which the high tide flooding frequency data are available and provides consistency between this information and relative SLR scenarios. The uncertainty in

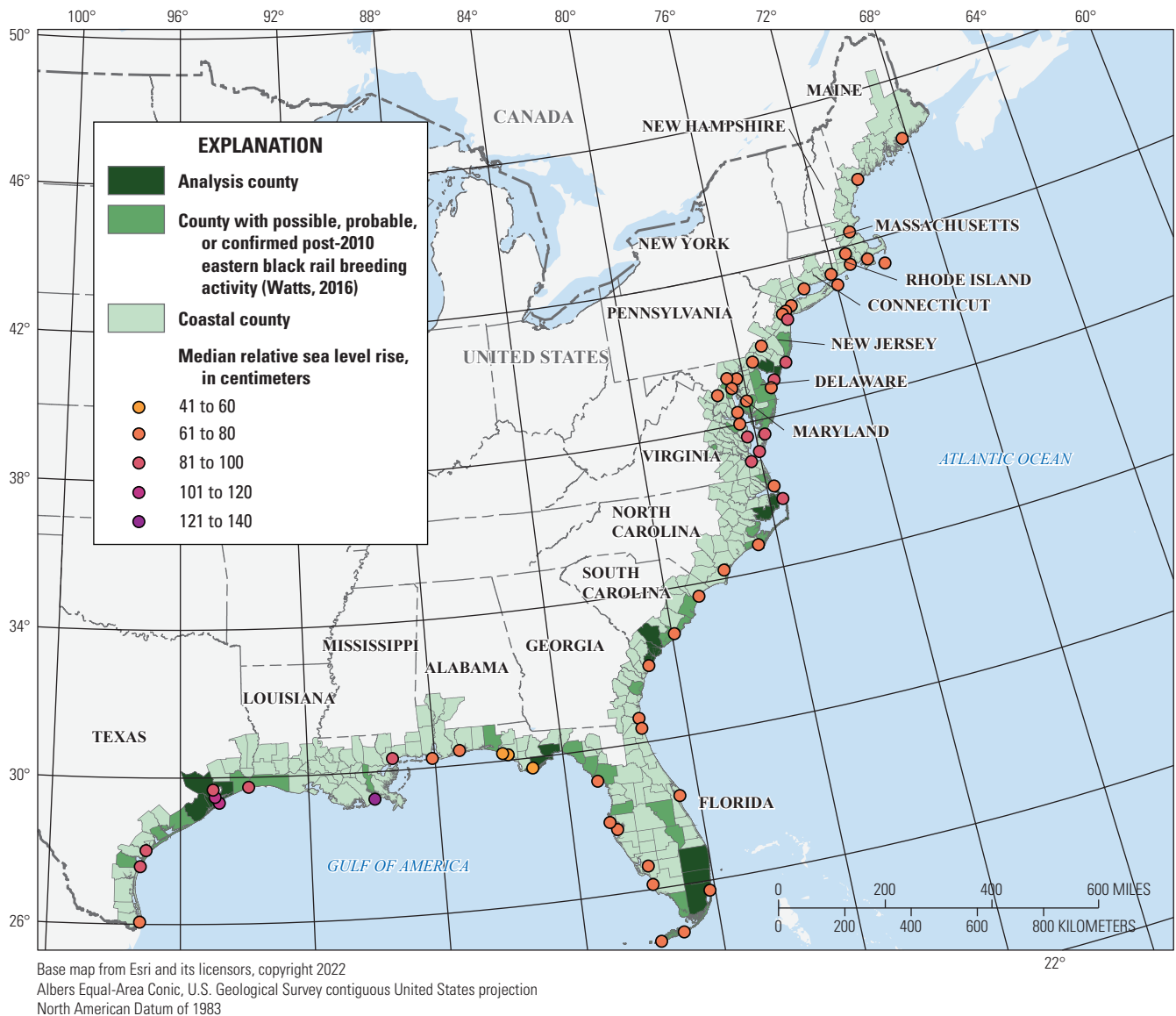


Figure 3. Map showing median projected relative sea level rise (in centimeters; relative to 2000 Mean Higher High Water) at tidal gage locations in 2100 under the intermediate low sea level rise scenario (0.5-meter global mean sea level rise by 2100). Values for the 17th and 83d percentile sea level rise projections at each tidal gage are available in [appendix 1](#).

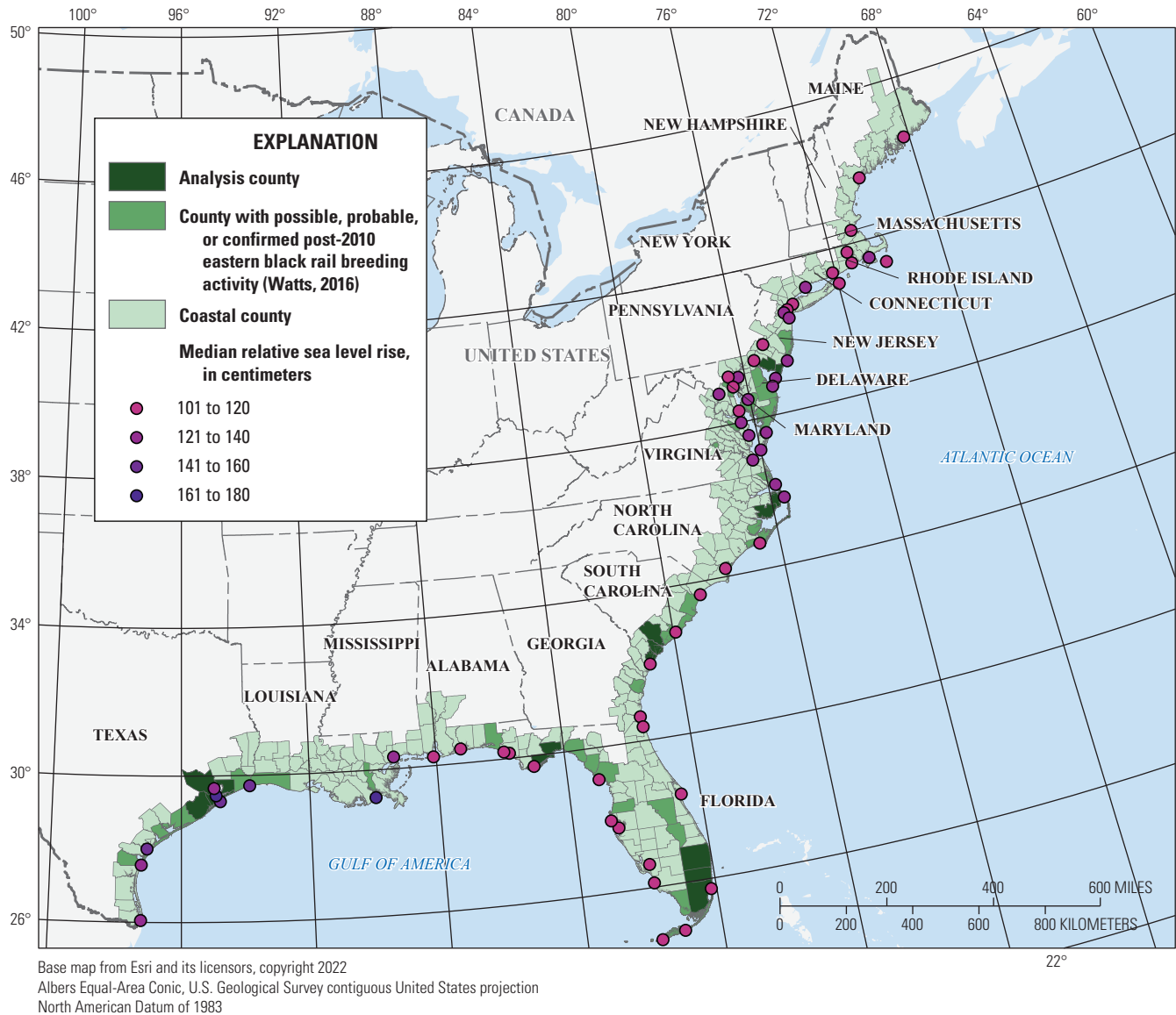


Figure 4. Map showing median projected relative sea level rise (in centimeters; relative to 2000 Mean Higher High Water) at tidal gage locations in 2100 under the intermediate sea level rise scenario (1-meter global mean sea level rise by 2100). Values for the 17th and 83d percentile sea level rise projections at each tidal gage are available in [appendix 1](#).

relative sea level projections (19–52 centimeters [cm] between 17th and 83d percentile projections at 2100 depending on scenario and gage) is greater than that between the tidal gage relative sea level and closest centroid relative sea level at several locations ([app. 1, table 1.2](#)). Additionally, the NOAA Sea Level Rise Viewer recommends that users round to the nearest 1-ft increment when investigating SLR inputs because of the elevation and tidal surface measurement accuracy (National Oceanic and Atmospheric Administration, 2022).

As sea level rises, the occurrence of high tide flooding events will increase by a factor of 5 to 10 from 2020 to 2050 under the intermediate SLR scenario in all locations along the Atlantic and U.S. Gulf coasts (Sweet and others, 2018; May and others, 2023). This increase is concerning for eastern

black rail given its vulnerability to flooding events during nesting and molting. High tides are driven by several factors including tidal cycles and will be altered by rising sea levels, changes in ocean circulation, and shifting wind patterns (Sweet and others, 2018). Year-to-year variability is expected given the uncertainty in these driving factors and tidal cycles and may result in rapid increases in high tide events and clusters of extreme events (Thompson and others, 2021). High tide flooding thresholds presented here are NOAA-derived thresholds based on a comparative analysis of tidal data across the United States (Sweet and others, 2018). High tide flooding thresholds are defined relative to their expected effect: minor (more disruptive than damaging), moderate (damaging), or major (destructive). Sweet and others (2018) developed a

spatially consistent threshold: 0.5 ± 0.19 m, 0.8 ± 0.25 m, and 1.17 ± 0.39 m for the three levels, respectively. Local variation of these thresholds is provided in [appendix 1](#).

We examined the projected number of high tide flooding days at these local thresholds developed by Thompson and others (2021) between April and September (a total of 183 days) to assess the changing risk to the eastern black rail during the breeding season (Thompson and others, 2021). The median number of projected high tide flooding events during the breeding season (a sum of the median projected value for each month) for the minor, moderate, and major high tide flooding thresholds is presented at each of the eastern black

rail gage locations under a range of SLR scenarios and time periods ([figs. 5, 6, 7, and 8](#)). Values bounding the very likely range of projected days are provided in [appendix 1](#) ([tables 1.3, 1.4, 1.5, 1.6, 1.7, and 1.8](#)). Note that these uncertainty values do not include the uncertainty within the SLR scenarios themselves (Thompson and others, 2021). These data are available in the Flooding Analysis Tool (https://sealevel.nasa.gov/data_tools/15/), developed in collaboration with the National Aeronautics and Space Administration Sea Level Change Team and the University of Hawaii Sea Level Center using the 2022 relative SLR scenarios (Sweet and others, 2022) and methodology from Thompson and others (2021).

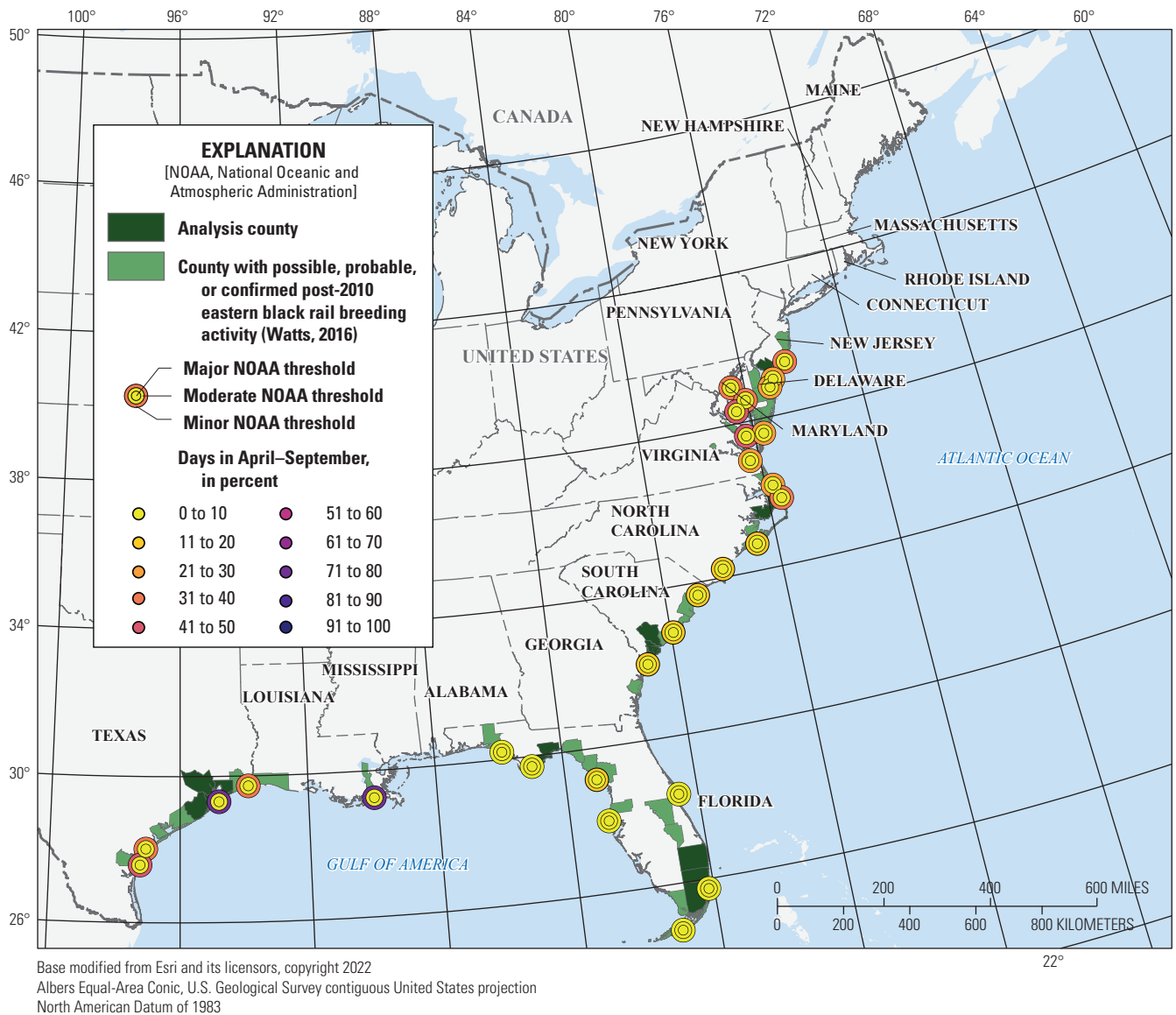


Figure 5. Map showing median projected number of high tide flooding events from April to September relative to local thresholds experienced at the minor (outer circle), moderate (middle circle), and major (center) National Oceanic and Atmospheric Administration-derived thresholds for the intermediate low sea level rise scenario at 2050. Values are presented at tidal gages closest to *Laterallus jamaicensis jamaicensis* (eastern black rail) breeding parishes and counties (Watts, 2016). Note that the median value is the sum of 50th-percentile projected flooding days for each individual month.

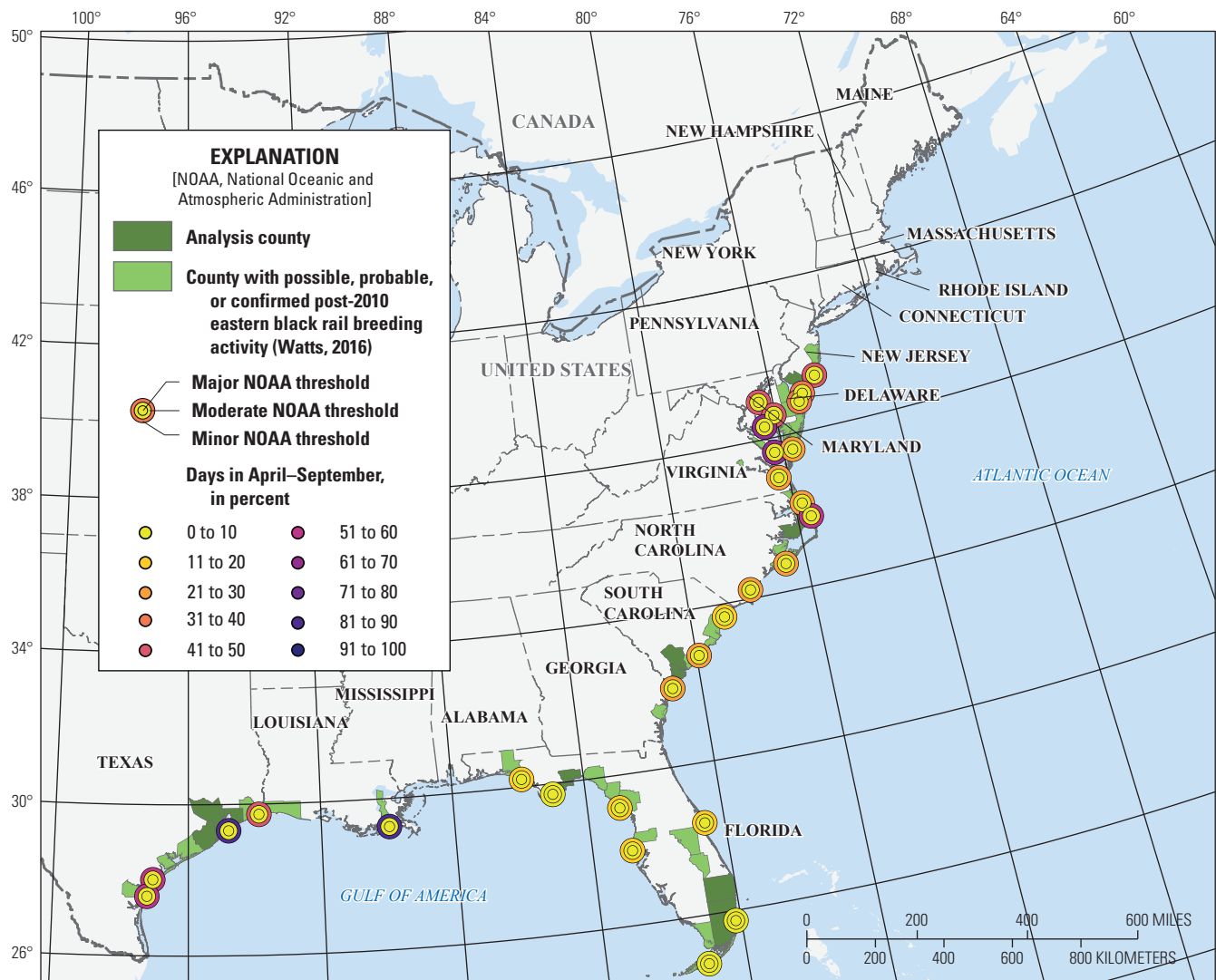


Figure 6. Map showing median projected number of high tide flooding events from April to September relative to local thresholds experienced at the minor (outer circle), moderate (middle circle), and major (center) National Oceanic and Atmospheric Administration-derived thresholds for the intermediate sea level rise scenario at 2050. Values are presented at tidal gages closest to *Laterallus jamaicensis jamaicensis* (eastern black rail) breeding parishes and counties (Watts, 2016). Note that the median value is the sum of 50th-percentile projected flooding days for each individual month.

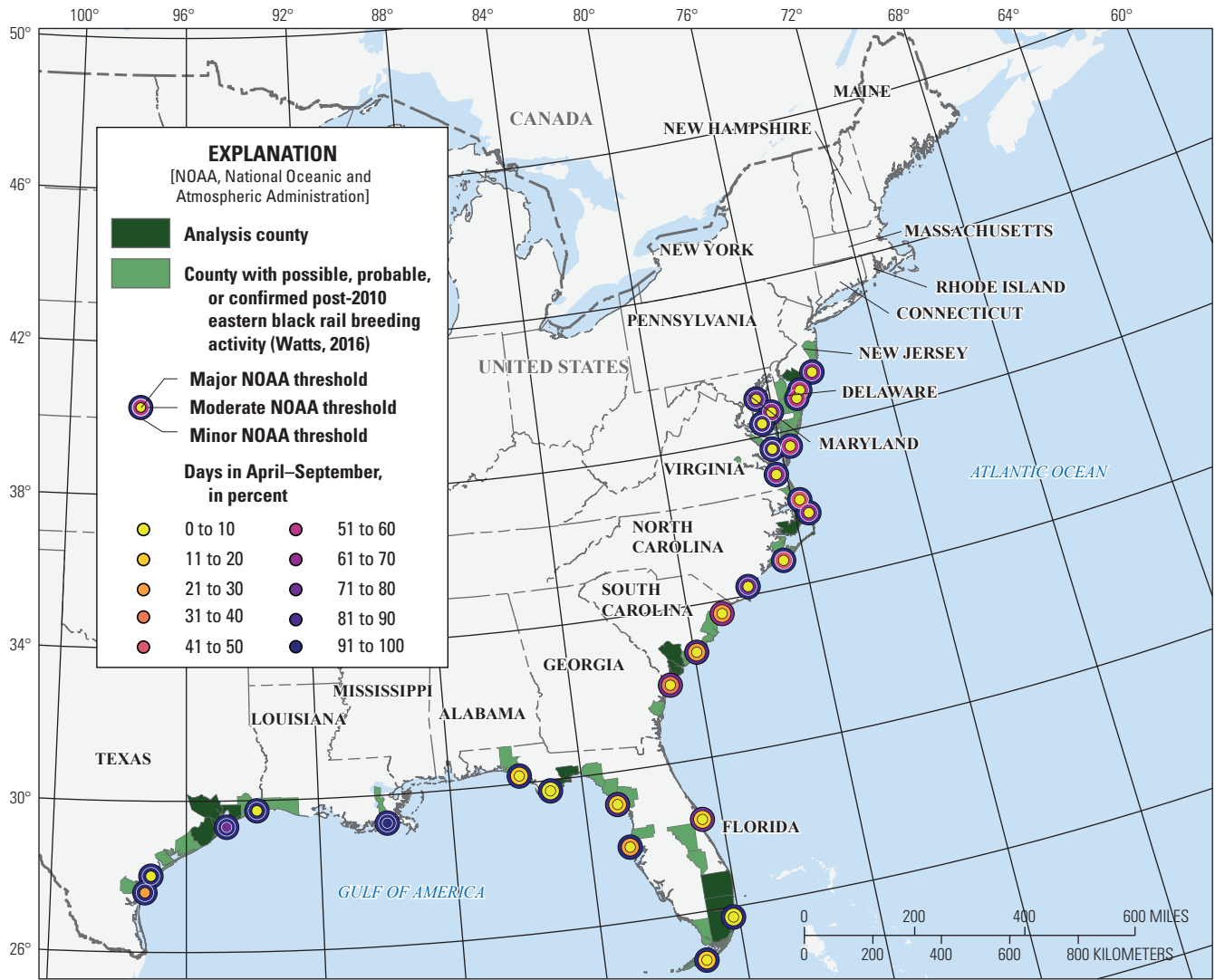


Figure 7. Map showing median projected number of high tide flooding events from April to September relative to local thresholds experienced at the minor (outer circle), moderate (middle circle), and major (center) National Oceanic and Atmospheric Administration-derived thresholds for the intermediate low sea level rise scenario at 2100. Values are presented at tidal gages closest to *Laterallus jamaicensis jamaicensis* (eastern black rail) breeding parishes and counties (Watts, 2016). Note that the median value is the sum of 50th-percentile projected flooding days for each individual month.

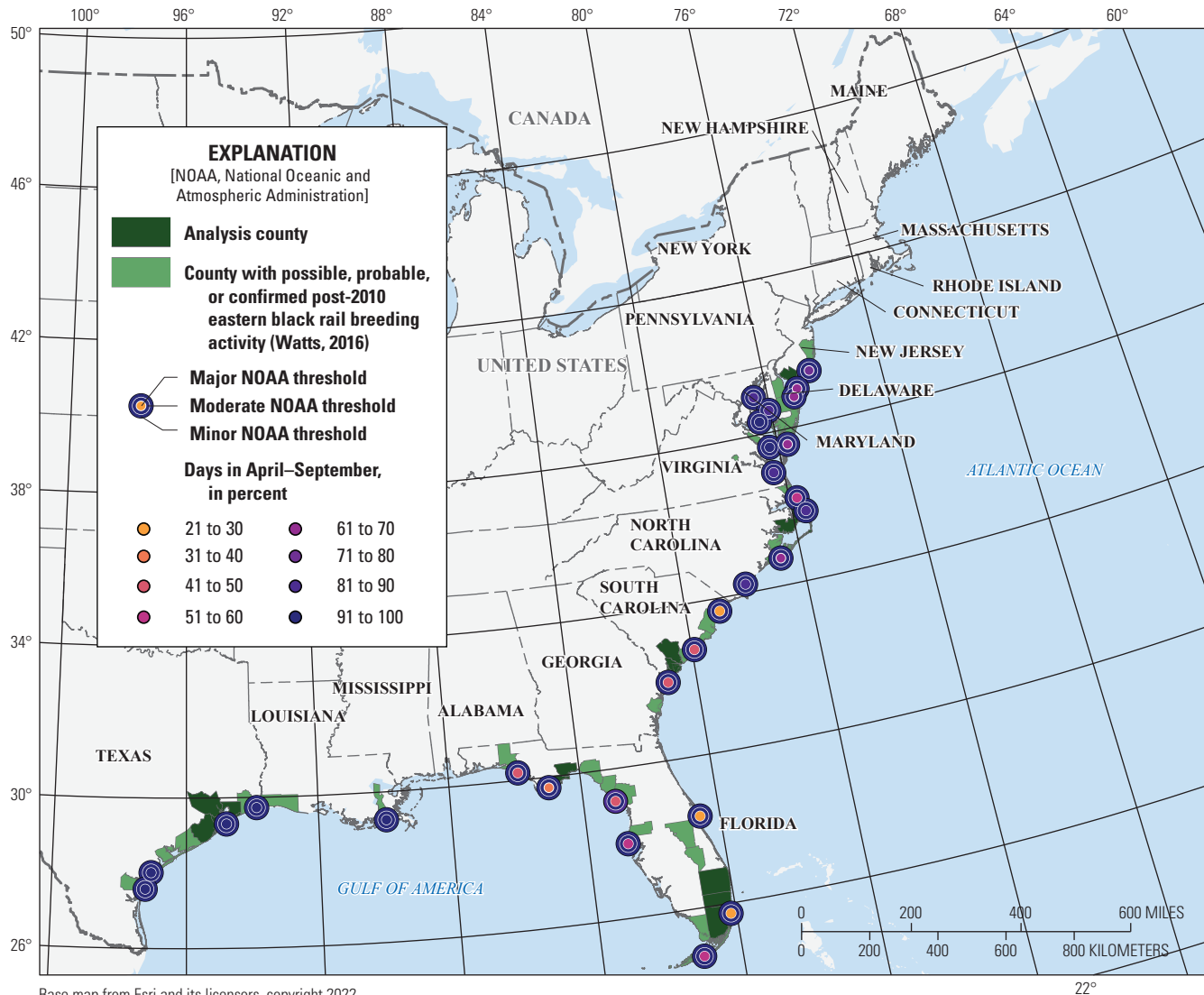


Figure 8. Map showing median projected number of high tide flooding events from April to September relative to local thresholds experienced at the minor (outer circle), moderate (middle circle), and major (center) National Oceanic and Atmospheric Administration-derived thresholds for the intermediate sea level rise scenario at 2100. Values are presented at tidal gages closest to *Laterallus jamaicensis jamaicensis* (eastern black rail) breeding parishes and counties (Watts, 2016). Note that the median value is the sum of 50th-percentile projected flooding days for each individual month.

Marsh Resilience

The inundation analysis in this report assumes static distributions of marsh land cover from historical data products. In reality, marsh landscapes are projected to respond and adapt to climate-driven changes in a variety of ways (FitzGerald and Hughes, 2019). To assess potential changes in marsh availability because of SLR, we examine two marsh products. First is a relative marsh resilience score from the tidal marsh resilience to sea level rise analysis developed by the NOAA Office for Coastal Management and the National Estuarine Research Reserve System (Stevens and others,

2023). This score assesses drainage basins with estuarine marsh (based on C-CAP data) according to the current marsh characteristics (using 2010 data), vulnerability to SLR, and adaptive capacity. These rankings do not provide an objective assessment of marsh health and resilience (the drainage basins are ranked from 1 to 10 relative to each other), but they do allow assessment across the full coastal distribution of the eastern black rail (fig. 9).

The marsh migration data developed by the NOAA Office of Coastal Management and available in the NOAA Sea Level Rise Viewer is used as a component of adaptive capacity in the marsh resilience score described previously. This marsh

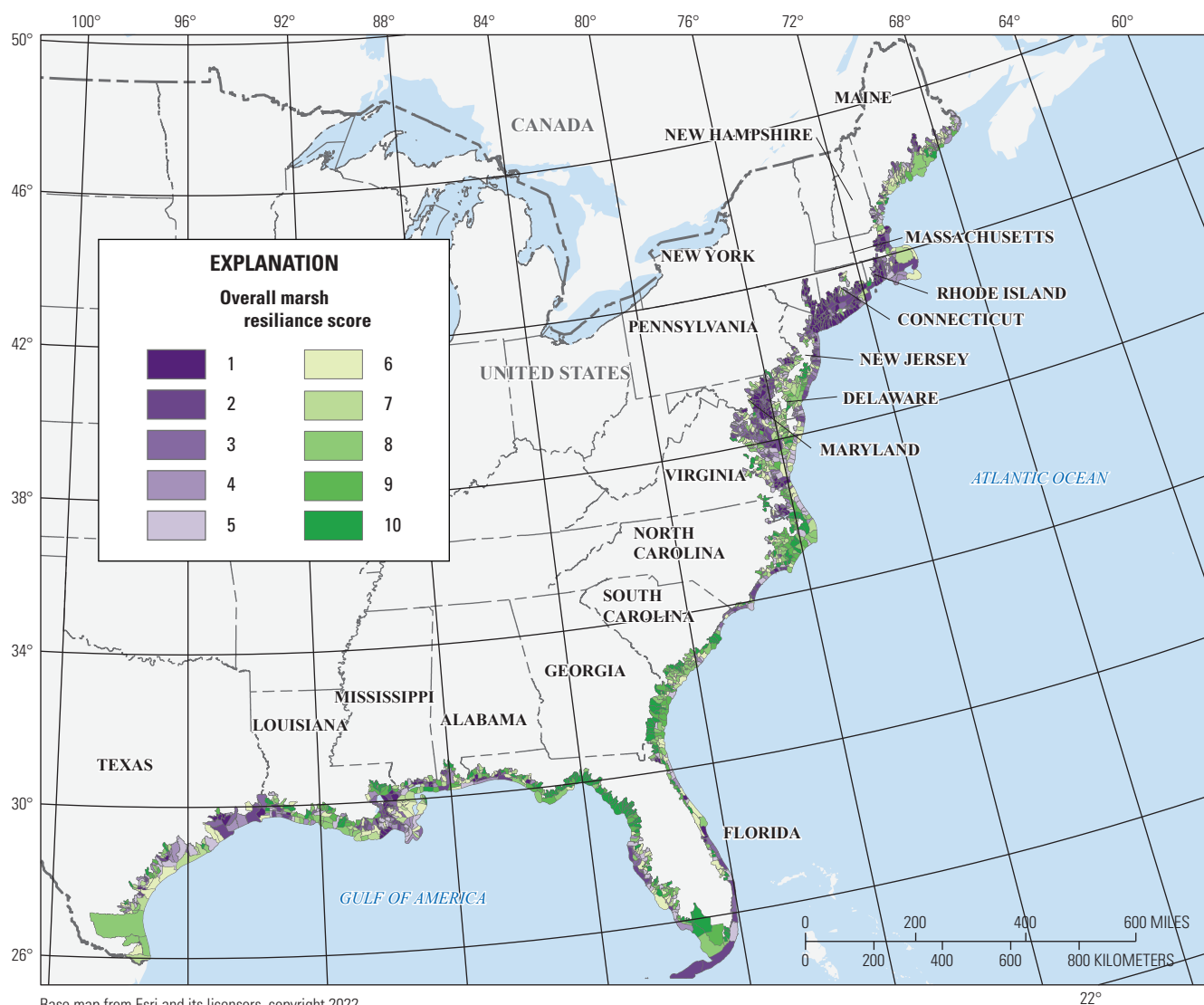


Figure 9. Map showing relative total marsh resilience scores developed by the National Oceanic and Atmospheric Administration Office for Coastal Management and the National Estuarine Research Reserve System (Stevens and others, 2023). Relative marsh resilience increases with score.

migration dataset can be analyzed in isolation to give a second assessment of potential marsh migration rates under multiple SLR heights (app. 2). In the absence of localized numerical marsh modeling for the full range of eastern black rail, these products allow for a comparison of relative marsh resilience. However, they do not capture local factors such as sediment dynamics in the same way as a more detailed model (Fuller and others, 2011) and do not account for competing pressures like mangrove expansion (Bardou and others, 2023).

Eastern Black Rail Coastal Flooding Risk Under Sea Level Rise

Rising sea levels leading to more frequent and more extensive high tide flood events will lead to a larger area of potential eastern black rail habitat inundation. To compare the consequences of these two changes, we assess the percentage of inundated habitat areas defined using the NLCD dataset under 0–9 ft of SLR for representative counties in six distributed regions (figs. 10, 11, 12, 13, 14, and 15). These percentages can be contextualized by comparing to the two sets of thresholds discussed here: relative SLR in figures 3 and 4 and in appendix 1 (tables 1.1–1.2; figures 1.1–1.2) and high

tide flood thresholds for each region, which are marked with dotted lines in figures 10, 11, 12, 13, 14, and 15 and defined in appendix 1 (tables 1.3, 1.4, 1.5, 1.6, 1.7, and 1.8).

Inundation levels are rounded to the nearest foot, and this inundation analysis quantifies the percentage of wetland habitat affected. Note that, depending on the amount of initial marsh habitat, a county with larger inundation percentages may still have larger absolute areas of a given habitat available. In most cases, the inundation of the transitional zone is at or below the inundation of the more inclusive emergent herbaceous wetland area because of its distance inland.

Combining the drivers (relative sea level rise and high tide flooding), exposure (inundation), and adaptive capacity (marsh resilience) allows for an assessment of relative risk from high tide flooding across the coastal breeding range of the eastern black rail. For example, the counties analyzed in Texas will experience a higher rate of relative SLR than other analysis areas (figs. 3 and 4) and more frequent occurrence of high tide flooding events (figs. 5, 6, 7, and 8), but they will initially experience a relatively low level of habitat inundation, likely because of the location and elevation of marsh areas along the Texas coast (fig. 10). This area has a medium marsh resilience score (fig. 9) and the potential for additional marsh area (app. 2).

The counties analyzed in New Jersey will likely experience a relatively high level of SLR (figs. 3 and 4) and high tide flooding events (figs. 5, 6, 7, and 8) and a steep increase in the percentage of habitat inundated under even small increases in relative sea level (fig. 11). This inundation is coupled with lower relative marsh resilience (fig. 9) despite potential area for marsh to expand (app. 2).

The northern coast of North Carolina will experience a relatively high level of SLR (figs. 3 and 4) and high tide flood event frequency (figs. 5, 6, 7, and 8; similar to the counties in New Jersey). Although habitat inundation is initially low in North Carolina under 0–2 ft of SLR, there is a steep increase under higher rates (fig. 12). The multifactor analysis of marsh resilience described previously gives this area a relatively good score (fig. 9); however, an analysis of potential marsh expansion area alone indicates little room for movement (app. 2).

The South Carolina counties exhibit a similar steep increase in coastal wetland inundation to the counties in New Jersey (fig. 13) but lower relative SLR (figs. 3 and 4) and high tide flooding event frequency than some other regions like North Carolina, New Jersey, and Texas (figs. 5, 6, 7, and 8). This area has a high marsh resilience score according to Stevens and others (2023); however, migration analysis indicates the potential for expansion only under low levels of SLR with loss at higher levels (app. 2).

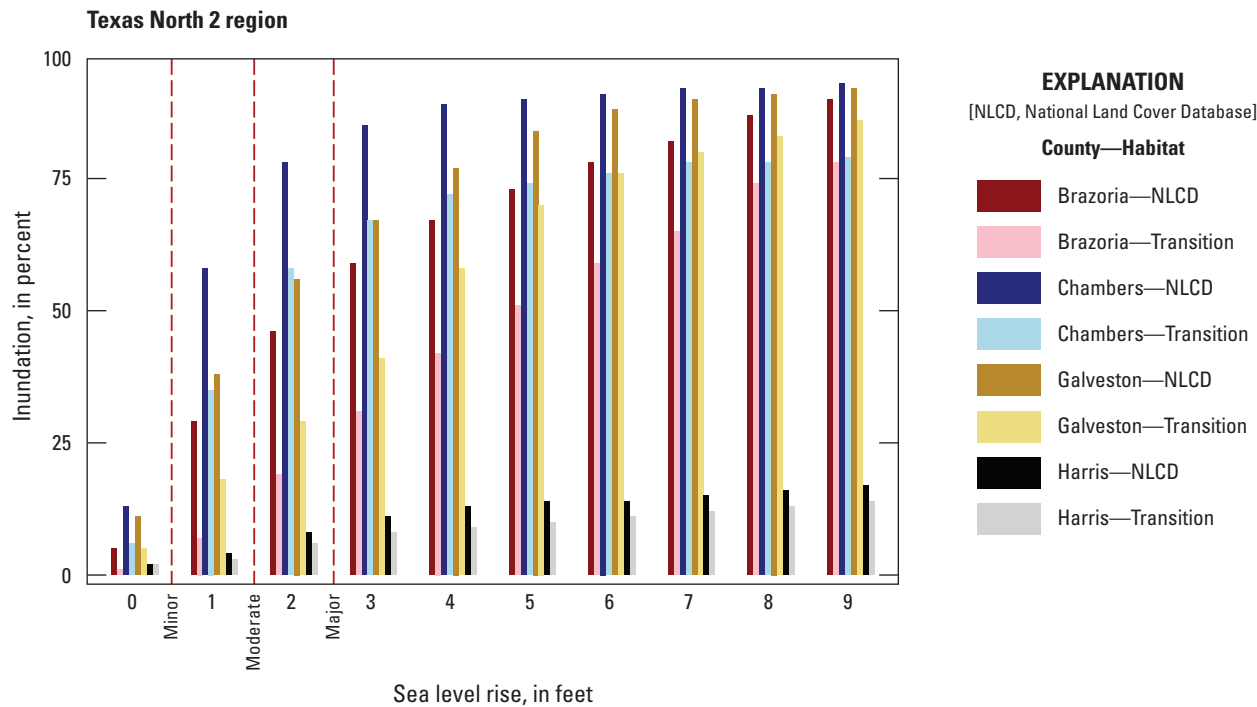


Figure 10. Graph showing percentage inundation under 0 to 9 feet of sea level rise above the historical tidal epoch Mean Higher High Water for analysis counties in the Texas North 2 region (app. 1, table 1.9). Habitat designations are National Land Cover Database (overall emergent herbaceous wetland) and transition (500-meter strip of emergent herbaceous wetland adjacent to woody wetland). Minor, moderate, and major lines denote the historical high tide flooding thresholds at the nearest tidal gage.

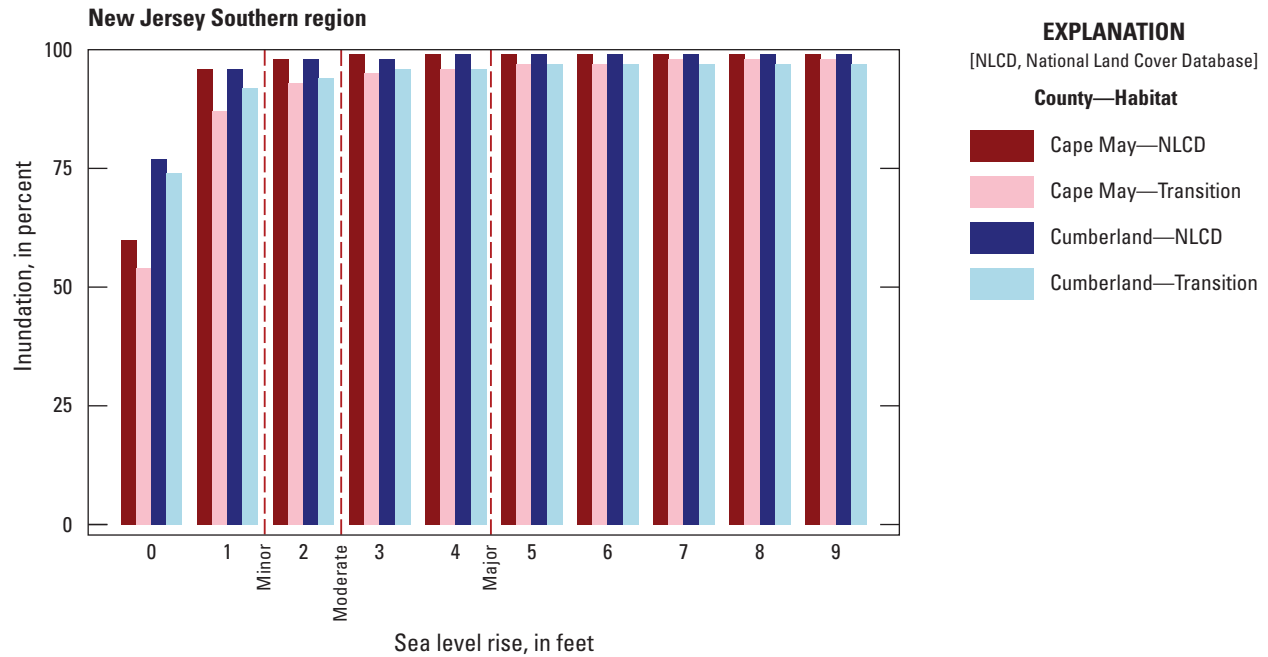


Figure 11. Graph showing percentage inundation under 0 to 9 feet of sea level rise above the historical tidal epoch Mean Higher High Water for analysis counties in the New Jersey Southern region (app. 1, table 1.9). Habitat designations are National Land Cover Database (overall emergent herbaceous wetland) and transition (500-meter strip of emergent herbaceous wetland adjacent to woody wetland). Minor, moderate, and major lines denote the historical high tide flooding thresholds at the nearest tidal gage.

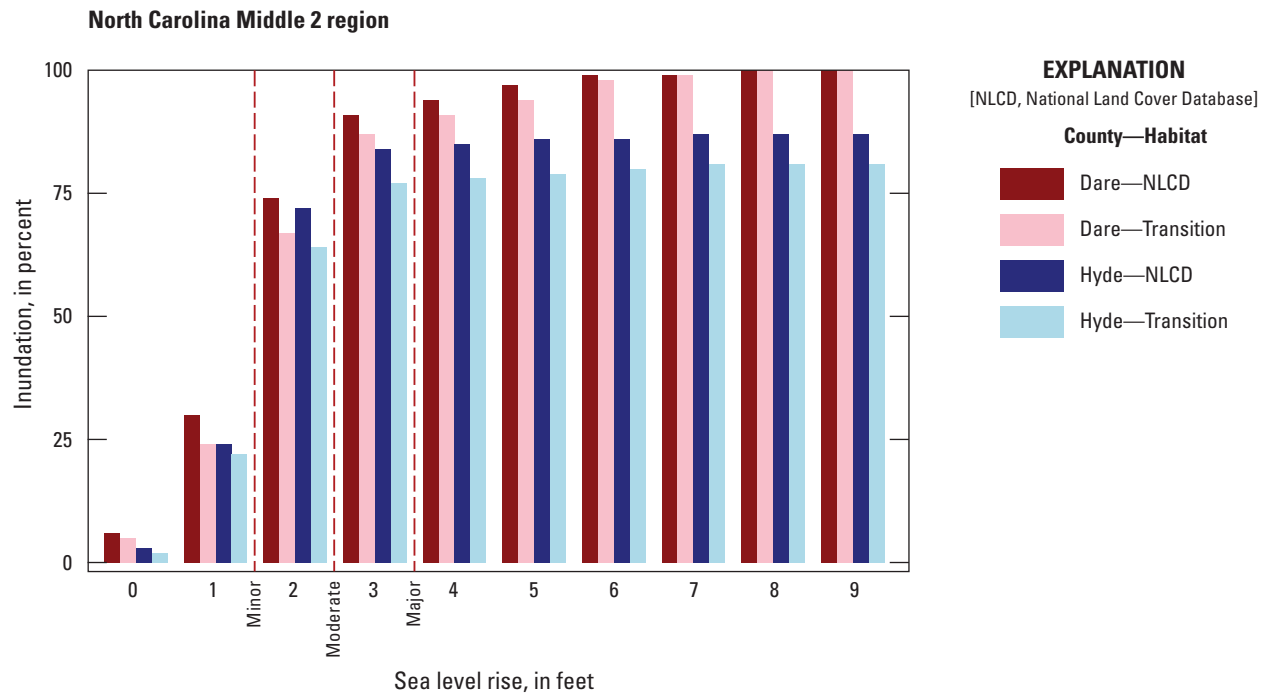


Figure 12. Graph showing percentage inundation under 0 to 9 feet of sea level rise above the historical tidal epoch Mean Higher High Water for analysis counties in the North Carolina Middle 2 region (app. 1, table 1.9). Habitat designations are National Land Cover Database (overall emergent herbaceous wetland) and transition (500-meter strip of emergent herbaceous wetland adjacent to woody wetland). Minor, moderate, and major lines denote the historical high tide flooding thresholds at the nearest tidal gage.

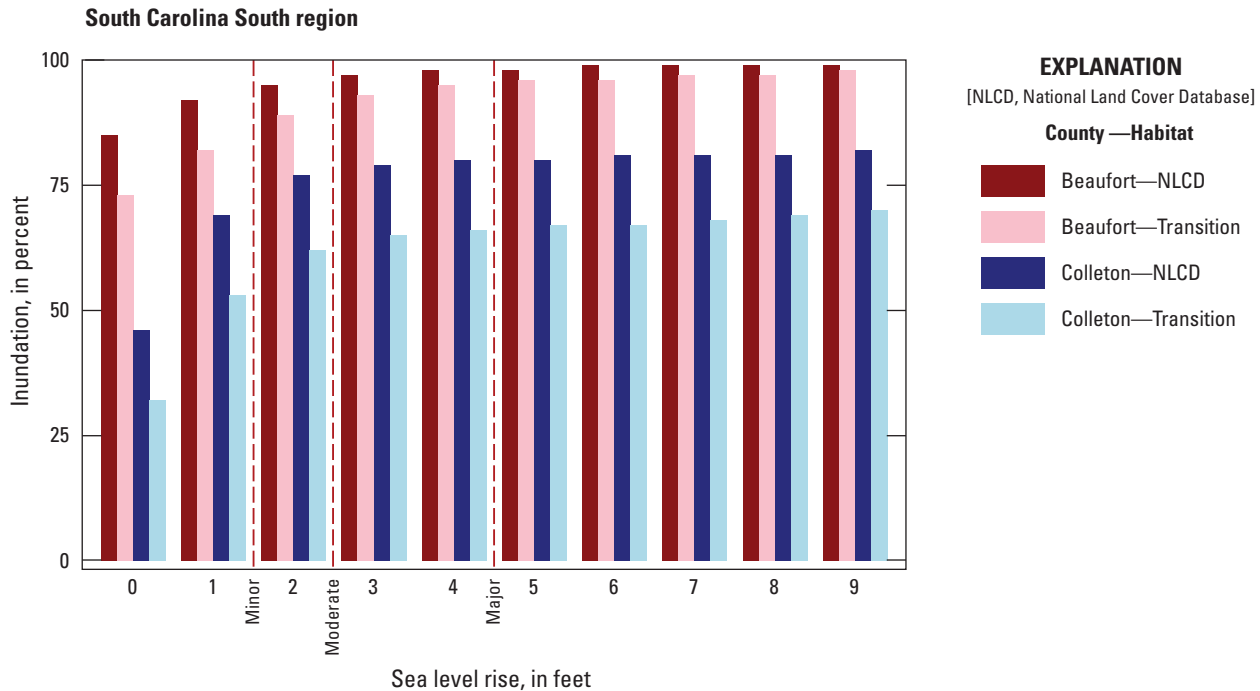


Figure 13. Graph showing percentage inundation under 0 to 9 feet of sea level rise above the historical tidal epoch Mean Higher High Water for analysis counties in the South Carolina South region (app. 1, table 1.9). Habitat designations are National Land Cover Database (overall emergent herbaceous wetland) and transition (500-meter strip of emergent herbaceous wetland adjacent to woody wetland). Minor, moderate, and major lines denote the historical high tide flooding thresholds at the nearest tidal gage.

The counties analyzed in the two regions of Florida are projected to experience the lowest relative SLR by 2100 (figs. 3 and 4) along with comparatively fewer (though still substantially increasing) high tide flooding events (figs. 5, 6, 7, and 8). The eastern Florida Panhandle will see a high percentage of habitat inundation under low to moderate levels of SLR (0–4 ft; fig. 14). This region also has a high marsh resilience (fig. 9) and higher potential marsh migration area (app. 2). The southeastern coast of the Florida Peninsula faces the most unique situation of the regions, mainly because of dense existing development along the coast and marsh areas limited to interior areas of the counties. Projected emergent herbaceous wetland habitat inundation is steadily increasing with SLR in the case of Miami-Dade County, whereas Broward County is only affected at higher levels of SLR and Palm Beach County faces no inundation of historical emergent herbaceous wetland in our analysis because these areas are farther inland (fig. 15). Marsh resilience scores for these counties are medium but are constrained to the immediate coast (fig. 9), and a countywide analysis indicates potential for estuarine and brackish marsh migration in Broward and Palm Beach Counties (app. 2).

By the end of century (2100), the high tide flooding frequency projections indicate daily occurrence of high tide events at or above the historical minor threshold under the intermediate low and intermediate SLR scenarios for all

coastal locations. Under the intermediate SLR scenario, the projected relative SLR in locations along the coast will meet or exceed even the major high tide flood threshold for most of the breeding season. The high tide flooding frequency data provide no insight about the maximum tidal level expected beyond this threshold and therefore provide little information about the true inundation levels experienced during extreme high tidal events in conjunction with SLR. For example, under 4 ft of relative SLR (the median projected SLR at 2100 under the intermediate scenario), a high tide flood event that contributes an additional 4 ft of inundation (an approximate major high tide flood threshold) would potentially contribute to a combined 8 ft of sea level depth above Mean Higher High Water. The analysis presented here can capture these events as meeting or exceeding the major high tide threshold of 4 ft but does not fully capture the habitat effect of the event. Therefore, analysis presented here includes potential SLR inundation amounts that exceed the historical high tide flooding thresholds and the projected base level of relative SLR, although the habitat percentages account only for historical wetland distribution. Additionally, storm surge and wave events will increase this inundation even further.

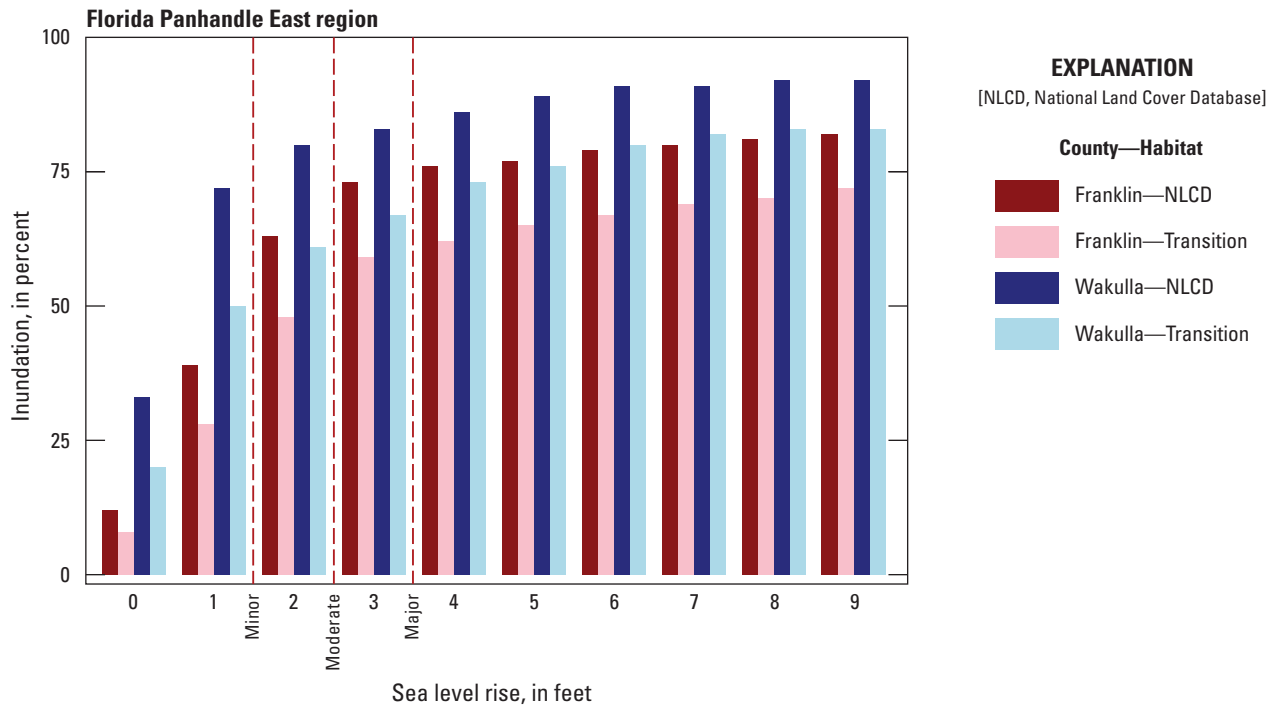


Figure 14. Graph showing percentage inundation under 0 to 9 feet of sea level rise above the historical tidal epoch Mean Higher High Water for analysis counties in the Florida Panhandle East region (app. 1, table 1.9). Habitat designations are National Land Cover Database (overall emergent herbaceous wetland) and transition (500-meter strip of emergent herbaceous wetland adjacent to woody wetland). Minor, moderate, and major lines denote the historical high tide flooding thresholds at the nearest tidal gage.

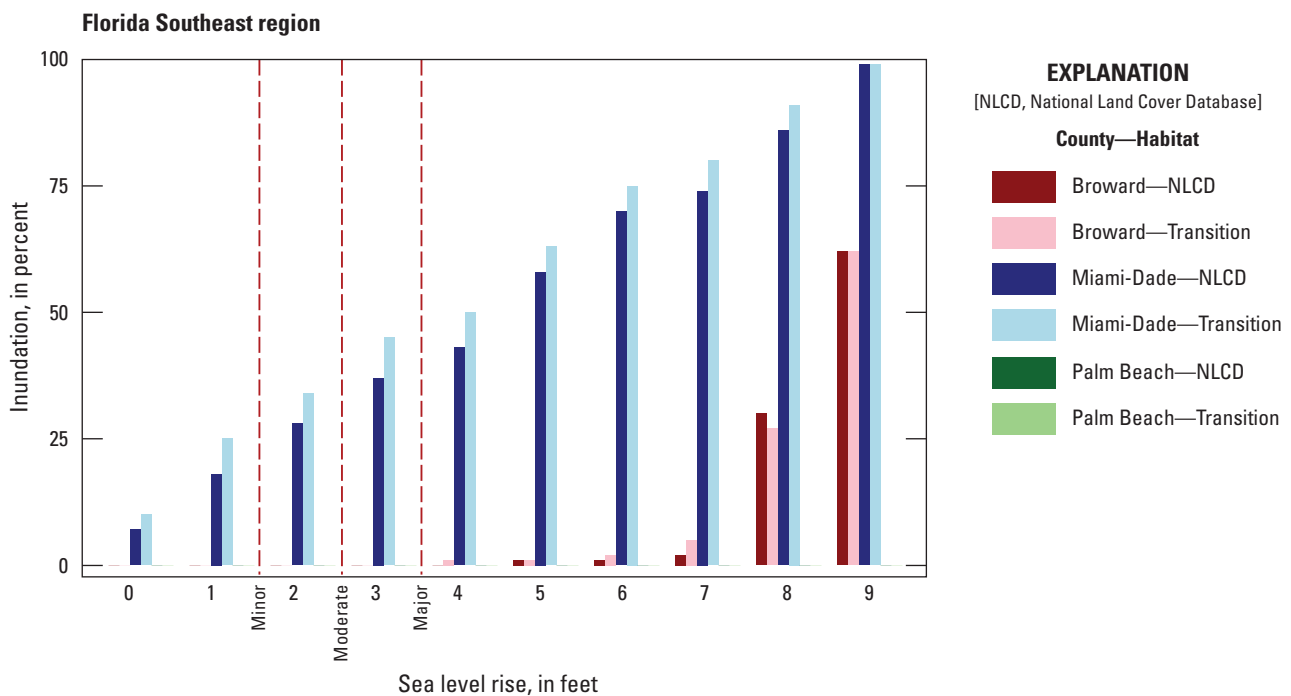


Figure 15. Graph showing percentage inundation under 0 to 9 feet of sea level rise above the historical tidal epoch Mean Higher High Water for analysis counties in the Florida Southeast region (app. 1, table 1.9). Habitat designations are National Land Cover Database (overall emergent herbaceous wetland) and transition (500-meter strip of emergent herbaceous wetland adjacent to woody wetland). Note that no habitat is inundated under the current definitions in Palm Beach County. Minor, moderate, and major lines denote the historical high tide flooding thresholds at the nearest tidal gage.

Adaptive Capacity

Eastern black rail traits may allow the subspecies to adapt to some of the effects of SLR and high tide flooding. Habitat analyses in this report rely on coarse habitat designation; the wetland and transition zone filters used to designate potential nesting areas in this report likely include lower marsh areas that eastern black rails would not select. Additionally, the 3-m resolution digital elevation models used to calculate SLR extent smooth out high points, eliminating some elevational heterogeneity, and have an average vertical root mean squared error of as much as 10 cm (National Oceanic and Atmospheric Administration, 2022). Nest site selection on elevated sites even 6 cm above the substrate (Legare and Eddleman 2001) or building nests on top of vegetation provides some additional distance from flooding; however, the rate of SLR and future high tide flooding are likely to exceed the elevational flood protection of many historical nest sites.

Latitudinal differences in breeding phenology are not evident (Watts, 2021); however, eastern black rails have been recorded to renest in a single season after a first failed attempt (Legare and Eddleman, 2001). The success rate and propensity to renest after nest loss are not known for the eastern black rail. Where quantified in *Anas platyrhynchos* (mallards; Linnaeus, 1758; Arnold and others, 2010) and *Charadrius melodus* (piping plover; Ord, 1824; Swift and others, 2020), reproductive success often decreases with multiple nest attempts in a year. This decrease in success is partially explained by generally lower success for later season nesting.

The eastern black rail does not only nest in the tidal wetlands that are the focus of this report but also in grassy fields and freshwater wetlands with no tidal influence and managed impoundments (Watts, 2016). SLR will have little to no effect on noncoastal wetlands, so they were excluded from this report that focuses on threats from saltwater inundation. Noncoastal wetlands may become increasingly important breeding areas as sea level rises, which has led some working groups to strategize creation of new nontidal, freshwater wetland habitat (Atlantic Coast Joint Venture, 2020; Watts and Beisler, 2021). Where studied, eastern black rail seems to preferentially use managed impoundments versus other wetlands (Roach and Barrett, 2015). Although we do not focus on coastal impoundments, they are largely captured as emergent herbaceous wetland in the NLCD 2021 dataset (Dewitz, 2023) used to delineate possible habitat. Low-lying hydrologically unconnected areas are captured in the SLR spatial data used in this report (National Oceanic and Atmospheric Administration, 2024). As inundation levels exceed barriers, these areas are inundated, but their representation depends on the detail available in the elevation data used to map inundation depth (National Oceanic and Atmospheric Administration, 2022).

This report does not consider the active role that management could play in protecting low elevation habitat or altering vegetation type and density in support of eastern black rail habitat needs and nesting in impoundments. These

management approaches could be a method for maintaining or creating habitat for eastern black rails and other marsh species in the future (Roach and Barrett, 2015). Areas like the Florida Everglades may provide additional benefits for marsh resilience and nesting availability through broader landscape-scale restoration and conservation efforts.

Compounding Stressors and Marsh Migration

Human modifications to coastal and near coastal systems are likely to impede marsh migration and marsh resilience without careful planning; for example, coastal squeeze between modified landscapes and rising sea level is likely to inhibit coastal habitat migration, and this risk is largest around developed, densely populated coastal areas (Borchert and others, 2018). In our analysis, the potential ability of wetlands to migrate is captured using the tidal marsh resilience to sea level rise analysis developed by the NOAA Office for Coastal Management and the National Estuarine Research Reserve System (Stevens and others, 2023) and marsh migration data available in the NOAA Sea Level Rise Viewer (National Oceanic and Atmospheric Administration, 2024). However, these products incorporate only historical patterns of urbanization and agriculture and do not include how potential area available for marsh migration will change with future human development. Projected marsh distribution shifts will also be slow and variable across the coast as inundation and salinity changes modify these ecosystems over time (Field and others, 2016).

This report details thresholds and changes associated with SLR and high tide flooding events, but storm surge and inland flooding from heavy rainfall events can also endanger nesting sites. In several locations along the coast, historical surge events of 5–10 ft have been detected since 1880 (Needham and Keim, 2012). This report does not catalog historical surge events or project how they might change; however, potential surge inundation can be mapped with the inundation values presented in [figures 10, 11, 12, 13, 14, and 15](#). Given the potential for renesting, this analysis focuses on the threat posed by increasingly frequent breeding season high tide flooding events driven in large part by SLR. Location-specific storm surge event frequency has not been projected given the uncertainty of landfalling hurricane frequency changes in the United States under various climate scenarios (Marvel and others, 2023). Inland freshwater flooding may pose issues to managed and unmanaged areas but is beyond the scope of this report, which focuses on the threat posed by saltwater inundation.

Conclusion

Laterallus jamaicensis jamaicensis (eastern black rail) are at greatest inundation risk during April–September when they inhabit coastal marsh areas for breeding and molting. They require wetland habitat for nesting but are unable to easily escape high water during breeding season. Some of these areas are currently affected by high tide flooding events, and the total area of coastal marsh expected to be affected by high tide flood events will increase because of SLR. This report investigates the change in days above historical high tide flood thresholds within this essential habitat for nesting eastern black rail. This approach does not capture some important processes to understanding the full effect to eastern black rail (such as fine scale microtopography variations, marsh evolution, coastal surge events, and management choices) because of data limitations regarding nesting preferences and habitat use. However, this study provides a qualitative assessment of coastal marsh inundation risk under a progression of SLR heights and incorporates the increasing frequency of days at or exceeding historical high tide flood thresholds across the eastern black rail's coastal breeding range.

Parishes and counties across the eastern black rail breeding range will face variable risks from high tide flooding and relative SLR. Eastern black rail breeding habitat is more at risk in areas with less historical habitat, higher rates of relative SLR, and a larger area of low elevation habitat. In some parishes and counties, coastal habitat has already been lost to land use change, reflecting the historical habitat loss that led to current subspecies decline. The distribution of habitat across the coast and the elevational gradient of the region are important factors for determining the risk to SLR and high tide flooding for eastern black rail. The resilience of marsh systems and the potential for inland marsh migration, which is mediated by current and future land development, are key considerations for future breeding habitat availability for these and other coastal wetland species. Lastly, the assessment of historical habitat inundation may underestimate potential nesting habitat if management options are not modeled and considered in future occupancy estimates.

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Appendix 1. Relative Sea Level Rise, High Tide Flooding Event Frequency, and Inundation Percentages at Tidal Gages and Analysis Counties

Figures 1.1 and 1.2 show the projected relative sea level rise at gage locations for the year 2050 to complement the projections shown in the main text for 2100. The main text figures and figures 1.1 and 1.2 show projections of relative sea level rise and high tide flooding days based on median projections. To maintain important information about the full range of projections data are presented in the following tables. Table 1.1 and 1.2 provide projections in centimeters for the low, median, and high projections of relative sea level rise at 2050 and 2100 respectively. Tables 1.3, 1.4, 1.5, 1.6, 1.7, and 1.8 provide the low, median and high percentiles of projected high tide flooding days at the minor, moderate, and major historical high tide flooding thresholds at 2050 and 2100. Table 1.9 details the change in habitat inundation between 0 and 5 feet of sea level rise and between 5 and 9 feet of sea level rise for each analysis county.

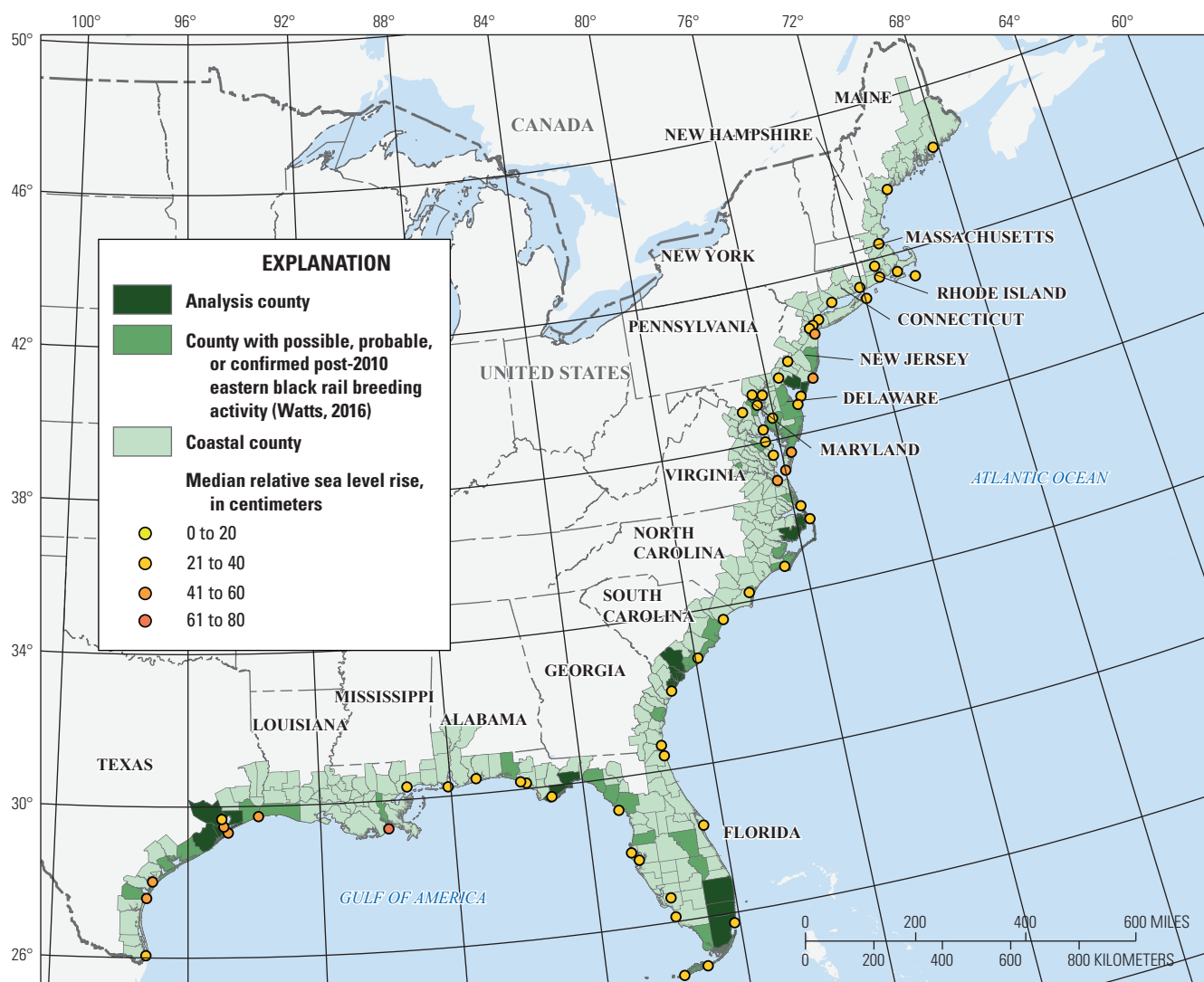


Figure 1.1. Map showing median projected relative sea level rise (in centimeters; relative to 2000 Mean Higher High Water) at tidal gage locations in 2050 under the intermediate low sea level rise scenario (0.5-meter global mean sea level rise by 2100). Values for the 17th and 83d percentile sea level rise projections at each tidal gage are available in [table 1.1](#).

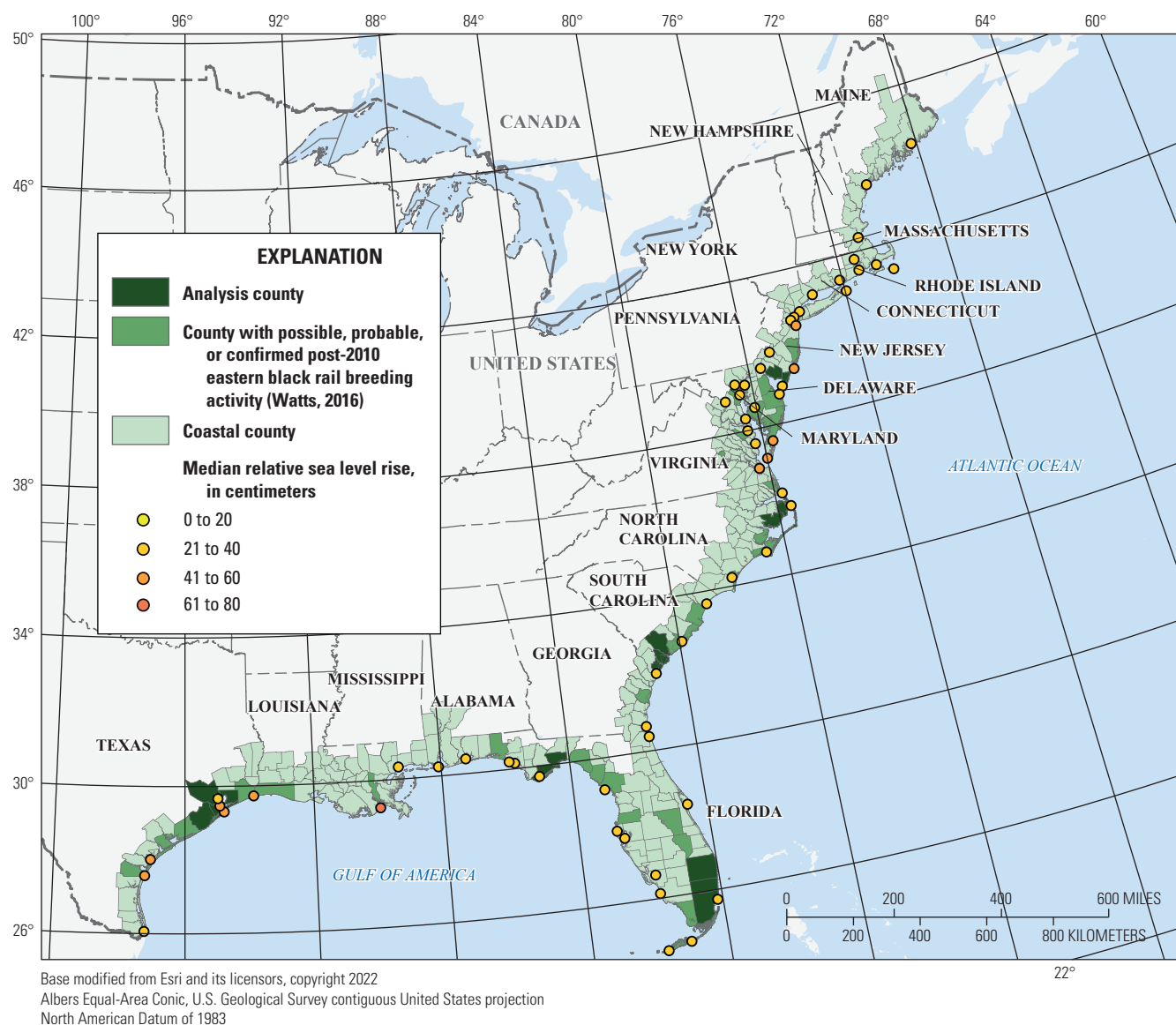


Figure 1.2. Map showing median projected relative sea level rise (in centimeters; relative to 2000 Mean Higher High Water) at tidal gage locations in 2050 under the intermediate sea level rise scenario (1-meter global mean sea level rise by 2100). Values for the 17th and 83d percentile sea level rise projections at each tidal gage are available in [table 1.1](#).

Table 1.1. Relative sea level rise projections under the intermediate low (0.5-meter global mean sea level rise by 2100) and intermediate sea level rise scenarios (1-meter global mean sea level rise by 2100) at 2050 at all tidal gage locations (fig. 1 in the main report; non-*Laterallus jamaicensis jamaicensis* [eastern black rail] gage and eastern black rail gage). Levels listed are the 17th (low), 50th (median), and 83d (high) percentiles of projected levels in the given scenario (Sweet and others, 2022).

[int., intermediate; cm, centimeter; med, median; --, no data or not applicable]

Tidal gage	Latitude	Longitude	2050 int. low (cm; low)	2050 int. low (cm; med)	2050 int. low (cm; high)	2050 int. (cm; low)	2050 int. (cm; med)	2050 int. (cm; high)
Bar Harbor	44.39	-68.20	24	33	43	28	37	47
Portland	43.66	-70.24	22	31	41	26	35	45
Boston	42.35	-71.05	26	35	44	29	39	49
Nantucket Island	41.29	-70.10	27	36	46	31	40	50
Woods Hole	41.52	-70.67	29	38	47	32	42	52
Newport	41.50	-71.33	26	36	45	30	39	50
Providence	41.81	-71.40	25	34	44	28	38	48
New London	41.37	-72.10	26	36	45	30	39	50
Montauk	41.05	-71.96	26	35	44	30	39	49
Bridgeport	41.17	-73.18	28	38	48	32	42	52
Kings Point	40.81	-73.77	27	36	45	30	39	49
The Battery	40.70	-74.01	28	37	46	32	40	50
Bergen Point	40.64	-74.15	29	38	47	32	41	51
Sandy Hook	40.47	-74.01	32	41	50	36	44	54
Atlantic City	39.36	-74.42	33	41	50	36	44	54
Cape May	38.97	-74.96	32	40	49	35	44	53
Philadelphia	39.93	-75.14	28	37	45	32	40	50
Reedy Point	39.56	-75.57	29	37	46	32	41	50
Lewes	38.78	-75.12	31	39	48	34	42	52
Tolchester Beach	39.21	-76.24	31	39	48	34	42	52
Baltimore	39.27	-76.58	30	38	47	33	41	51
Annapolis	38.98	-76.48	29	37	46	32	41	50
Cambridge	38.57	-76.06	31	39	47	34	42	52
Washington D.C.	38.87	-77.02	32	40	48	34	43	53
Solomons Island	38.32	-76.45	29	38	46	32	41	50
Lewisetta	38.00	-76.46	31	39	48	34	43	52
Windmill Point	37.62	-76.29	--	40	--	--	44	--
Wachapreague	37.60	-75.68	--	41	--	--	45	--
Kiptopeke	37.17	-75.99	33	42	50	36	45	55
Sewells Point	36.95	-76.33	34	42	51	37	46	55
Duck	36.18	-75.75	32	40	48	35	43	53
Oregon Inlet	35.80	-75.55	32	40	48	35	44	53
Beaufort	34.72	-76.67	28	35	43	31	39	48
Wilmington	34.23	-77.95	25	32	40	28	36	46
Springmaid Pier	33.66	-78.92	25	32	39	28	36	45
Charleston	32.78	-79.92	28	35	42	32	39	47
Fort Pulaski	32.04	-80.90	29	35	42	32	39	47
Fernandina Beach	30.67	-81.47	25	31	37	28	35	43
Mayport	30.39	-81.43	25	32	38	29	36	44

24 Potential Effects of Sea Level Rise and High Tide Flooding on *Laterallus jamaicensis jamaicensis*

Table 1.1. Relative sea level rise projections under the intermediate low (0.5-meter global mean sea level rise by 2100) and intermediate sea level rise scenarios (1-meter global mean sea level rise by 2100) at 2050 at all tidal gage locations (fig. 1 in the main report; non-*Laterallus jamaicensis jamaicensis* [eastern black rail] gage and eastern black rail gage). Levels listed are the 17th (low), 50th (median), and 83d (high) percentiles of projected levels in the given scenario (Sweet and others, 2022).—Continued

[int., intermediate; cm, centimeter; med, median; --, no data or not applicable]

Tidal gage	Latitude	Longitude	2050 int. low (cm; low)	2050 int. low (cm; med)	2050 int. low (cm; high)	2050 int. (cm; low)	2050 int. (cm; med)	2050 int. (cm; high)
Trident Pier	28.42	−80.59	24	30	36	28	34	42
Virginia Key	25.73	−80.16	24	30	36	28	34	42
Vaca Key	24.71	−81.11	25	31	37	28	35	43
Key West	24.55	−81.81	25	30	36	27	34	42
Naples	26.13	−81.81	24	30	37	27	34	43
Fort Meyers	26.65	−81.87	24	30	36	27	34	42
St. Petersburg	27.76	−82.63	26	32	38	29	36	44
Clearwater	27.98	−82.83	26	32	38	29	36	44
Cedar Key	29.14	−83.03	23	29	36	26	33	42
Apalachicola	29.72	−84.98	22	29	35	25	32	41
Panama City	30.15	−85.70	22	28	35	25	32	40
Panama City Beach	30.20	−85.87	--	28	--	--	32	--
Pensacola	30.40	−87.21	23	30	36	27	34	42
Dauphin Island	30.25	−88.08	28	34	41	31	38	46
Bay Waveland	30.33	−89.33	34	40	47	37	44	52
Grand Isle	29.26	−89.96	54	61	67	57	64	73
Sabine Pass	29.73	−93.87	40	47	54	43	50	59
Galveston Pier 21	29.31	−94.79	44	50	57	47	54	63
Eagle Point	29.47	−94.92	--	57	--	--	60	--
Morgans Point	29.67	−94.98	--	39	--	--	43	--
Rockport	28.02	−97.05	41	47	54	43	51	59
Corpus Christi	27.58	−97.22	36	42	49	39	46	54
Port Isabel	26.06	−97.22	31	38	45	34	42	50

Table 1.2. Relative sea level rise projections under the intermediate low (0.5-meter of global mean sea level rise by 2100) and intermediate (1-meter of global mean sea level rise by 2100) sea level rise scenarios at 2100 at all tidal gage locations (fig. 1 in the main report; non-*Laterallus jamaicensis jamaicensis* [eastern black rail] gage and eastern black rail gage). Levels listed are the 17th (low), 50th (median), and 83d (high) percentiles of projected levels in the given scenario (Sweet and others, 2022).

[int., intermediate; cm, centimeter; med, median; --, no data or not applicable]

Tidal gage	Latitude	Longitude	2100 int. low (cm; low)	2100 int. low (cm; med)	2100 int. low (cm; high)	2100 int. (cm; low)	2100 int. (cm; med)	2100 int. (cm; high)
Bar Harbor	44.39	-68.20	50	66	85	83	111	135
Portland	43.66	-70.24	45	61	80	79	107	130
Boston	42.35	-71.05	54	69	88	88	115	137
Nantucket Island	41.29	-70.10	57	73	92	92	119	141
Woods Hole	41.52	-70.67	60	76	95	96	123	145
Newport	41.50	-71.33	56	71	90	91	117	140
Providence	41.81	-71.40	52	68	87	87	113	135
New London	41.37	-72.10	56	71	89	91	117	138
Montauk	41.05	-71.96	55	70	88	91	116	136
Bridgeport	41.17	-73.18	60	76	94	96	122	143
Kings Point	40.81	-73.77	57	71	88	92	116	136
The Battery	40.70	-74.01	60	74	91	95	119	139
Bergen Point	40.64	-74.15	61	76	93	97	121	141
Sandy Hook	40.47	-74.01	68	82	99	103	128	147
Atlantic City	39.36	-74.42	69	83	100	104	128	147
Cape May	38.97	-74.96	67	81	98	101	125	145
Philadelphia	39.93	-75.14	60	73	91	94	118	138
Reedy Point	39.56	-75.57	61	75	92	94	119	139
Lewes	38.78	-75.12	65	79	96	99	123	143
Tolchester Beach	39.21	-76.24	65	78	96	98	122	142
Baltimore	39.27	-76.58	63	76	93	96	120	140
Annapolis	38.98	-76.48	62	75	93	95	119	139
Cambridge	38.57	-76.06	65	78	95	98	122	142
Washington D.C.	38.87	-77.02	67	80	97	100	124	144
Solomons Island	38.32	-76.45	62	76	93	96	119	139
Lewisetta	38.00	-76.46	66	79	96	100	123	143
Windmill Point	37.62	-76.29	--	81	--	--	125	--
Wachapreague	37.60	-75.68	--	82	--	--	126	--
Kiptopeke	37.17	-75.99	71	84	101	104	128	148
Sewells Point	36.95	-76.33	73	86	102	107	130	149
Duck	36.18	-75.75	68	80	97	102	125	144
Oregon Inlet	35.80	-75.55	69	81	97	104	126	144
Beaufort	34.72	-76.67	61	72	85	96	118	134
Wilmington	34.23	-77.95	55	66	80	90	111	128
Springmaid Pier	33.66	-78.92	55	66	79	90	112	128
Charleston	32.78	-79.92	62	72	85	97	118	134
Fort Pulaski	32.04	-80.90	63	73	85	98	119	134
Fernandina Beach	30.67	-81.47	54	64	77	90	111	127
Mayport	30.39	-81.43	56	66	78	91	112	128

Table 1.2. Relative sea level rise projections under the intermediate low (0.5-meter of global mean sea level rise by 2100) and intermediate (1-meter of global mean sea level rise by 2100) sea level rise scenarios at 2100 at all tidal gage locations (fig. 1 in the main report; non-*Laterallus jamaicensis jamaicensis* [eastern black rail] gage and eastern black rail gage). Levels listed are the 17th (low), 50th (median), and 83d (high) percentiles of projected levels in the given scenario (Sweet and others, 2022).—Continued

[int., intermediate; cm, centimeter; med, median; --, no data or not applicable]

Tidal gage	Latitude	Longitude	2100 int. low (cm; low)	2100 int. low (cm; med)	2100 int. low (cm; high)	2100 int. (cm; low)	2100 int. (cm; med)	2100 int. (cm; high)
Trident Pier	28.42	–80.59	55	64	76	91	111	126
Virginia Key	25.73	–80.16	57	66	77	92	113	127
Vaca Key	24.71	–81.11	58	67	77	94	114	129
Key West	24.55	–81.81	56	65	76	92	113	127
Naples	26.13	–81.81	54	64	75	91	112	127
Fort Meyers	26.65	–81.87	55	64	76	91	112	127
St. Petersburg	27.76	–82.63	58	67	79	94	115	130
Clearwater	27.98	–82.83	58	67	79	94	115	130
Cedar Key	29.14	–83.03	52	62	74	88	108	124
Apalachicola	29.72	–84.98	50	60	72	85	106	122
Panama City	30.15	–85.70	49	59	71	84	105	121
Panama City Beach	30.20	–85.87	--	60	--	--	106	--
Pensacola	30.40	–87.21	52	63	75	87	108	125
Dauphin Island	30.25	–88.08	62	72	84	96	117	134
Bay Waveland	30.33	–89.33	75	85	98	108	130	147
Grand Isle	29.26	–89.96	116	126	138	149	171	189
Sabine Pass	29.73	–93.87	86	96	109	120	142	159
Galveston Pier 21	29.31	–94.79	94	104	116	128	150	167
Eagle Point	29.47	–94.92	--	115	--	--	161	--
Morgans Point	29.67	–94.98	--	87	--	--	131	--
Rockport	28.02	–97.05	87	97	110	121	143	160
Corpus Christi	27.58	–97.22	77	87	99	111	133	150
Port Isabel	26.06	–97.22	68	78	90	102	124	141

Table 1.3. Breeding season (April–September) minor high tide flooding days (maximum=183) under the intermediate low and intermediate sea level rise scenarios at 2050 at *Laterallus jamaicensis jamaicensis* (eastern black rail) gage locations (fig. 1 in the main report). Levels listed are about the 5th (low), 50th (median), and about the 95th (high) percentiles of projected levels in the given scenario (that is, the range that bounds the very likely [greater than 90 percent] projections). Seasonal values are calculated by summing the low, median, and high bounds for each individual month across the breeding season (Thompson and others, 2021; Sweet and others, 2022; National Aeronautics and Space Administration Sea Level Change Team, 2024).

[HTF, high tide flood; cm, centimeter; int., intermediate; med, median]

Tidal gage	Minor HTF threshold (cm)	Minor 2050 int. low (days; low)	Minor 2050 int. low (days; med)	Minor 2050 int. low (days; high)	Minor 2050 int. (days; low)	Minor 2050 int. (days; med)	Minor 2050 int. (days; high)
Atlantic City	56	22	63	112	31	75	124
Cape May	57	12	50	102	22	64	116
Lewes	56	9	44	97	15	56	110
Annapolis	52	17	66	131	28	87	150
Cambridge	52	17	66	126	31	86	146
Solomons Island	52	30	89	153	48	112	166
Windmill Point	53	28	88	157	45	113	168
Wachapreague	56	8	44	101	16	56	114
Sewells Point	53	13	54	119	15	69	139
Duck	55	9	42	92	15	51	106
Oregon Inlet	51	15	70	148	24	94	162
Beaufort	54	3	27	81	6	43	99
Wilmington	56	1	29	105	2	43	131
Springmaid Pier	57	5	23	64	7	32	75
Charleston	57	4	31	77	6	41	90
Fort Pulaski	59	6	35	83	10	46	93
Trident Pier	54	2	15	56	4	23	70
Virginia Key	52	0	1	39	4	18	58
Vaca Key	51	0	3	24	0	8	48
Clearwater	54	2	17	50	4	29	71
Cedar Key	55	2	22	63	5	32	79
Apalachicola	52	0	6	42	1	13	61
Panama City Beach	52	0	14	49	1	22	66
Grand Isle	43	73	146	181	105	163	183
Sabine Pass	52	9	61	146	16	78	162
Galveston Pier 21	52	52	134	181	70	153	183
Rockport	50	11	71	164	20	97	176
Corpus Christi	53	22	88	162	32	104	172

Table 1.4. Breeding season (April–September) minor high tide flooding days (maximum=183) under the intermediate low and intermediate sea level rise scenarios at 2100 at *Laterallus jamaicensis jamaicensis* (eastern black rail) gage locations (fig. 1 in the main report). Levels listed are about the 5th (low), 50th (median), and about the 95th (high) percentiles of projected levels in the given scenario (that is, the range that bounds the very likely [greater than 90 percent] projections). Seasonal values are calculated by summing the low, median, and high bounds for each individual month across the breeding season (Thompson and others, 2021; Sweet and others, 2022; National Aeronautics and Space Administration Sea Level Change Team, 2024).

[HTF, high tide flooding; cm, centimeter; int., intermediate; med, median]

Tidal gage	Minor HTF threshold (cm)	Minor 2100 int. low (days; low)	Minor 2100 int. low (days; med)	Minor 2100 int. low (days; high)	Minor 2100 int. (days; low)	Minor 2100 int. (days; med)	Minor 2100 int. (days; high)
Atlantic City	56	163	182	183	183	183	183
Cape May	57	157	180	183	183	183	183
Lewes	56	152	180	183	183	183	183
Annapolis	52	177	182	183	183	183	183
Cambridge	52	180	183	183	183	183	183
Solomons Island	52	182	183	183	183	183	183
Windmill Point	53	182	183	183	183	183	183
Wachapreague	56	155	181	183	183	183	183
Sewells Point	53	173	183	183	183	183	183
Duck	55	147	176	183	183	183	183
Oregon Inlet	51	181	183	183	183	183	183
Beaufort	54	156	182	183	183	183	183
Wilmington	56	176	183	183	183	183	183
Springmaid Pier	57	69	119	163	180	183	183
Charleston	57	96	148	179	183	183	183
Fort Pulaski	59	91	141	177	183	183	183
Trident Pier	54	70	131	172	183	183	183
Virginia Key	52	121	178	183	183	183	183
Vaca Key	51	172	183	183	183	183	183
Clearwater	54	144	175	183	183	183	183
Cedar Key	55	115	158	179	183	183	183
Apalachicola	52	116	172	183	183	183	183
Panama City Beach	52	147	177	183	183	183	183
Grand Isle	43	183	183	183	183	183	183
Sabine Pass	52	183	183	183	183	183	183
Galveston Pier 21	52	183	183	183	183	183	183
Rockport	50	183	183	183	183	183	183
Corpus Christi	53	183	183	183	183	183	183

Table 1.5. Breeding season (April–September) moderate high tide flooding days (maximum=183) under the intermediate low and intermediate sea level rise scenarios at 2050 at *Laterallus jamaicensis jamaicensis* (eastern black rail) gage locations (fig. 1 in the main report). Levels listed are about the 5th (low), 50th (median), and about the 95th (high) percentiles of projected levels in the given scenario (that is, the range that bounds the very likely [greater than 90 percent] projections). Seasonal values are calculated by summing the low, median, and high bounds for each individual month across the breeding season (Thompson and others, 2021; Sweet and others, 2022; National Aeronautics and Space Administration Sea Level Change Team, 2024).

[HTF, high tide flooding; cm, centimeter; int., intermediate; med, median]

Tidal gage	Moderate HTF threshold (cm)	Moderate 2050 int. low (days; low)	Moderate 2050 int. low (days; med)	Moderate 2050 int. low (days; high)	Moderate 2050 int. (days; low)	Moderate 2050 int. (days; med)	Moderate 2050 int. (days; high)
Atlantic City	84	0	3	24	0	5	30
Cape May	85	0	1	15	0	2	22
Lewes	84	0	0	13	0	1	18
Annapolis	81	0	0	10	0	1	15
Cambridge	82	0	0	6	0	1	9
Solomons Island	81	0	1	16	0	3	23
Windmill Point	83	0	3	16	0	5	22
Wachapreague	85	0	2	14	0	2	19
Sewells Point	82	0	2	17	0	4	21
Duck	83	0	2	13	0	3	20
Oregon Inlet	81	0	1	10	0	1	16
Beaufort	83	0	0	3	0	0	6
Wilmington	84	0	0	3	0	0	6
Springmaid Pier	85	0	0	8	0	1	14
Charleston	85	0	0	10	0	1	13
Fort Pulaski	87	0	1	15	0	1	21
Trident Pier	82	0	0	4	0	0	6
Virginia Key	81	0	0	1	0	0	2
Vaca Key	81	0	0	0	0	0	0
Clearwater	83	0	0	1	0	0	2
Cedar Key	84	0	0	4	0	1	7
Apalachicola	82	0	0	0	0	0	1
Panama City Beach	81	0	0	0	0	0	1
Grand Isle	72	0	6	38	0	13	57
Sabine Pass	82	0	0	14	0	2	23
Galveston Pier 21	81	0	10	56	0	16	73
Rockport	80	0	1	11	0	1	19
Corpus Christi	82	0	1	33	0	5	45

Table 1.6. Breeding season (April–September) moderate high tide flooding days (maximum=183) under the intermediate low and intermediate sea level rise scenarios at 2100 at *Laterallus jamaicensis jamaicensis* (eastern black rail) gage locations (fig. 1 in the main report). Levels listed are about the 5th (low), 50th (median), and about the 95th (high) percentiles of projected levels in the given scenario (that is, the range that bounds the very likely [greater than 90 percent] projections). Seasonal values are calculated by summing the low, median, and high bounds for each individual month across the breeding season (Thompson and others, 2021; Sweet and others, 2022; National Aeronautics and Space Administration Sea Level Change Team, 2024).

[HTF, high tide flooding; cm, centimeter; int., intermediate; med, median]

Tidal gage	Moderate HTF threshold (cm)	Moderate 2100 int. low (days; low)	Moderate 2100 int. low (days; med)	Moderate 2100 int. low (days; high)	Moderate 2100 int. (days; low)	Moderate 2100 int. (days; med)	Moderate 2100 int. (days; high)
Atlantic City	84	64	114	161	182	183	183
Cape May	85	49	101	151	180	183	183
Lewes	84	39	93	145	180	183	183
Annapolis	81	66	136	175	182	183	183
Cambridge	82	64	128	169	183	183	183
Solomons Island	81	104	161	181	183	183	183
Windmill Point	83	99	164	182	183	183	183
Wachapreague	85	39	94	152	179	183	183
Sewells Point	82	47	117	172	183	183	183
Duck	83	36	84	139	177	183	183
Oregon Inlet	81	39	124	177	183	183	183
Beaufort	83	21	76	137	183	183	183
Wilmington	84	48	138	183	183	183	183
Springmaid Pier	85	5	38	85	126	169	183
Charleston	85	12	54	108	156	180	183
Fort Pulaski	87	20	61	111	146	179	183
Trident Pier	82	2	26	73	146	179	183
Virginia Key	81	2	17	68	183	183	183
Vaca Key	81	2	19	77	183	183	183
Clearwater	83	11	44	90	181	183	183
Cedar Key	84	3	31	79	172	182	183
Apalachicola	82	0	14	65	177	183	183
Panama City Beach	81	7	44	101	181	183	183
Grand Isle	72	183	183	183	183	183	183
Sabine Pass	82	87	169	183	183	183	183
Galveston Pier 21	81	183	183	183	183	183	183
Rockport	80	130	182	183	183	183	183
Corpus Christi	82	148	183	183	183	183	183

Table 1.7. Breeding season (April–September) major high tide flooding days (maximum=183) under the intermediate low and intermediate sea level rise scenarios in 2050 at *Laterallus jamaicensis jamaicensis* (eastern black rail) gage locations (fig. 1 in the main report). Levels listed are about the 5th (low), 50th (median), and about the 95th (high) percentiles of projected levels in the given scenario (that is, the range that bounds the very likely [greater than 90 percent] projections). Seasonal values are calculated by summing the low, median, and high bounds for each individual month across the breeding season (Thompson and others, 2021; Sweet and others, 2022; National Aeronautics and Space Administration Sea Level Change Team, 2024).

[HTF, high tide flooding; cm, centimeter; int., intermediate; med, median]

Tidal gage	Major HTF threshold (cm)	Major 2050 int. low (days; low)	Major 2050 int. low (days; med)	Major 2050 int. low (days; high)	Major 2050 int. (days; low)	Major 2050 int. (days; med)	Major 2050 int. (days; high)
Atlantic City	123	0	0	0	0	0	0
Cape May	124	0	0	0	0	0	0
Lewes	123	0	0	0	0	0	0
Annapolis	119	0	0	0	0	0	0
Cambridge	120	0	0	0	0	0	0
Solomons Island	119	0	0	0	0	0	0
Windmill Point	120	0	0	0	0	0	0
Wachapreague	123	0	0	0	0	0	0
Sewells Point	120	0	0	0	0	0	0
Duck	122	0	0	0	0	0	0
Oregon Inlet	118	0	0	0	0	0	0
Beaufort	121	0	0	0	0	0	0
Wilmington	123	0	0	0	0	0	0
Springmaid Pier	124	0	0	0	0	0	0
Charleston	124	0	0	0	0	0	0
Fort Pulaski	126	0	0	0	0	0	0
Trident Pier	121	0	0	0	0	0	0
Virginia Key	119	0	0	0	0	0	0
Vaca Key	118	0	0	0	0	0	0
Clearwater	121	0	0	0	0	0	0
Cedar Key	122	0	0	0	0	0	0
Apalachicola	119	0	0	0	0	0	0
Panama City Beach	119	0	0	0	0	0	0
Grand Isle	110	0	0	0	0	0	0
Sabine Pass	119	0	0	0	0	0	0
Galveston Pier 21	119	0	0	0	0	0	0
Rockport	117	0	0	0	0	0	0
Corpus Christi	120	0	0	0	0	0	0

Table 1.8. Breeding season (April–September) major high tide flooding days (maximum=183) under the intermediate low and intermediate sea level rise scenarios in 2100 at *Laterallus jamaicensis jamaicensis* (eastern black rail) gage locations (fig. 1 in the main report). Levels listed are about the 5th (low), 50th (median), and about the 95th (high) percentiles of projected levels in the given scenario (that is, the range that bounds the very likely [greater than 90 percent] projections). Seasonal values are calculated by summing the low, median, and high bounds for each individual month across the breeding season (Thompson and others, 2021; Sweet and others, 2022; National Aeronautics and Space Administration Sea Level Change Team, 2024).

[HTF, high tide flooding; cm, centimeter; int., intermediate; med, median]

Tidal gage	Major HTF threshold (cm)	Major 2100 int. low (days; low)	Major 2100 int. low (days; med)	Major 2100 int. low (days; high)	Major 2100 int. (days; low)	Major 2100 int. (days; med)	Major 2100 int. (days; high)
Atlantic City	123	0	6	28	88	139	175
Cape May	124	0	2	20	71	125	169
Lewes	123	0	1	16	61	119	166
Annapolis	119	0	1	13	110	163	180
Cambridge	120	0	0	8	108	160	179
Solomons Island	119	0	3	22	145	177	182
Windmill Point	120	0	5	26	151	179	183
Wachapreague	123	0	3	20	59	124	171
Sewells Point	120	0	4	24	80	150	182
Duck	122	0	1	14	54	109	159
Oregon Inlet	118	0	1	12	81	158	182
Beaufort	121	0	0	4	51	117	169
Wilmington	123	0	0	19	81	165	183
Springmaid Pier	124	0	0	4	15	54	103
Charleston	124	0	0	6	26	74	130
Fort Pulaski	126	0	0	15	35	81	131
Trident Pier	121	0	0	0	8	48	103
Virginia Key	119	0	0	0	13	55	131
Vaca Key	118	0	0	0	33	99	166
Clearwater	121	0	0	0	44	94	144
Cedar Key	122	0	0	2	26	76	131
Apalachicola	119	0	0	0	3	40	116
Panama City Beach	119	0	0	0	30	85	145
Grand Isle	110	116	170	183	183	183	183
Sabine Pass	119	0	8	55	137	182	183
Galveston Pier 21	119	51	131	180	183	183	183
Rockport	117	0	9	66	174	183	183
Corpus Christi	120	2	41	118	174	183	183

Table 1.9. Analysis region and change in inundation percentage for analysis counties (fig. 1 in the main report). Columns list the change in inundation percentage among certain levels of sea level rise for habitat designations. For example, historical emergent herbaceous wetland in Cape May County, New Jersey, is 60 percent inundated under 0 feet (ft) of sea level rise relative to Mean Higher High Water and 99 percent inundated under 5 ft of sea level rise relative to Mean Higher High Water. This translates to an additional 39 percent of the historical emergent herbaceous wetland being inundated as sea level rise increases from 0 to 5 ft.

[%, percent; NLCD, National Land Cover Database; herb., herbaceous; ft, foot; SLR, sea level rise; NJ, New Jersey; NC, North Carolina; SC, South Carolina; FL, Florida; SE, Southeast; Pan, Panhandle; TX, Texas]

County	State	Post-2010 status (Watts, 2016)	<i>Laterallus jamaicensis jamaicensis</i> (eastern black rail) tidal gage	Analysis region	% NLCD emergent herb. inundation change 0–5 ft SLR	% NLCD emergent herb. inundation change 5–9 ft SLR	% NLCD emergent herb. transition inundation change 0–5 ft SLR	% NLCD emergent herb. transition inundation change 5–9 ft SLR
Cape May	NJ	Probable	Cape May	NJ Southern	39	0	43	1
Cumberland	NJ	Probable	Cape May	NJ Southern	22	0	23	0
Dare	NC	Probable	Oregon Inlet	NC Middle 2	91	3	89	6
Hyde	NC	Probable	Oregon Inlet	NC Middle 2	83	1	77	2
Beaufort	SC	Possible	Fort Pulaski	SC South	13	1	23	2
Colleton	SC	Confirmed	Charleston	SC South	34	2	35	3
Broward	FL	Probable	Virginia Key	FL SE	1	61	1	61
Franklin	FL	Probable	Apalachicola	FL Pan East	65	5	57	7
Miami-Dade	FL	Probable	Virginia Key	FL SE	51	41	53	36
Palm Beach	FL	Probable	Virginia Key	FL SE	0	0	0	0
Wakulla	FL	Probable	Apalachicola	FL Pan East	56	3	56	7
Brazoria	TX	Probable	Galveston Pier 21	TX North 2	68	17	50	27
Chambers	TX	Probable	Galveston Pier 21	TX North 2	77	3	68	5
Galveston	TX	Probable	Galveston Pier 21	TX North 2	73	8	65	16
Harris	TX	Possible	Galveston Pier 21	TX North 2	12	3	8	4

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Appendix 2. Marsh Migration Data Analysis

An analysis of marsh migration potential was developed by the National Oceanic and Atmospheric Administration (NOAA) Office of Coastal Management. This dataset projects marsh movement under sea level rise (SLR) amounts across the analysis counties in the absence of additional development (post-2005–6). This dataset uses marsh classifications from the NOAA Coastal Change Analysis Program dataset, and future marsh types are determined using an elevational gradient between the water line and upland areas (National Oceanic and Atmospheric Administration, 2017, 2022). Accretion rates of 2, 4, and 6 millimeters per year are presented as options in the NOAA Sea Level Rise Viewer and can be analyzed for a given location either to understand the range of potential changes or to align with observed historical rates. Wetland changes here are presented based on an absolute level of SLR and are therefore independent of an accretion choice. If analysis is done looking at potential inundation under a given SLR scenario and period, accretion should be considered to prevent overestimation.

Analysis of changes in estuarine, palustrine, and brackish/transitional marsh is provided here for absolute relative sea level, which may occur in a range of combined

SLR and accretion scenarios (figs. 2.1, 2.2, 2.3, 2.4, 2.5, and 2.6). All types of marsh are potential areas for *Laterallus jamaicensis jamaicensis* (eastern black rail) nesting. Values are presented as a percentage change in area relative to the amount of a given marsh type present under 0 feet (ft) of SLR. Note that brackish/transitional marsh is included in this product but is not a class in the original Coastal Change Analysis Program layer and therefore is not included in the distribution of land cover types at 0-ft SLR. The percentage change for this marsh type is relative to the area present under 1 ft of SLR and therefore does not have a data point at 0 ft of SLR.

This product can capture the potential area available for marsh migration given the topography and historical development of a county but does not indicate where marshes are likely to expand or be lost because of any other factors, including expansion of developed areas. This dataset also does not provide a sense of temporal scale for marsh movement and loss because of inundation or salinity effects. The transient nature of high tide flooding events does not align with the time scale of these long-term habitat changes, and therefore, they are not incorporated in the main analysis.

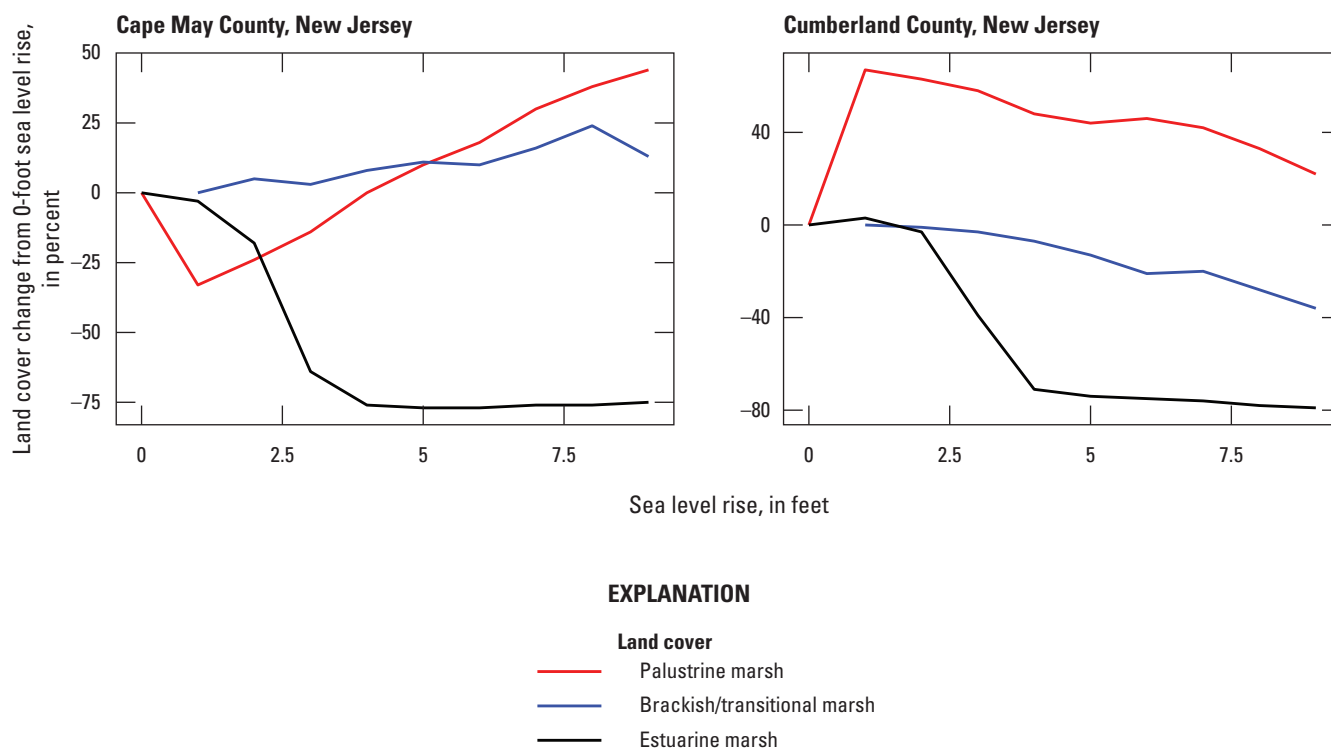


Figure 2.1. Graphs showing percentage change in area of palustrine marsh, brackish/transitional marsh, and estuarine marsh from 0 to 9 feet (ft) of sea level rise above Mean Higher High Water in analysis counties in the New Jersey Southern region (app. 1, table 1.9). Note that brackish/transitional marsh is not included in the base Coastal Change Analysis Program dataset and therefore has no area in the 0-ft mapping.

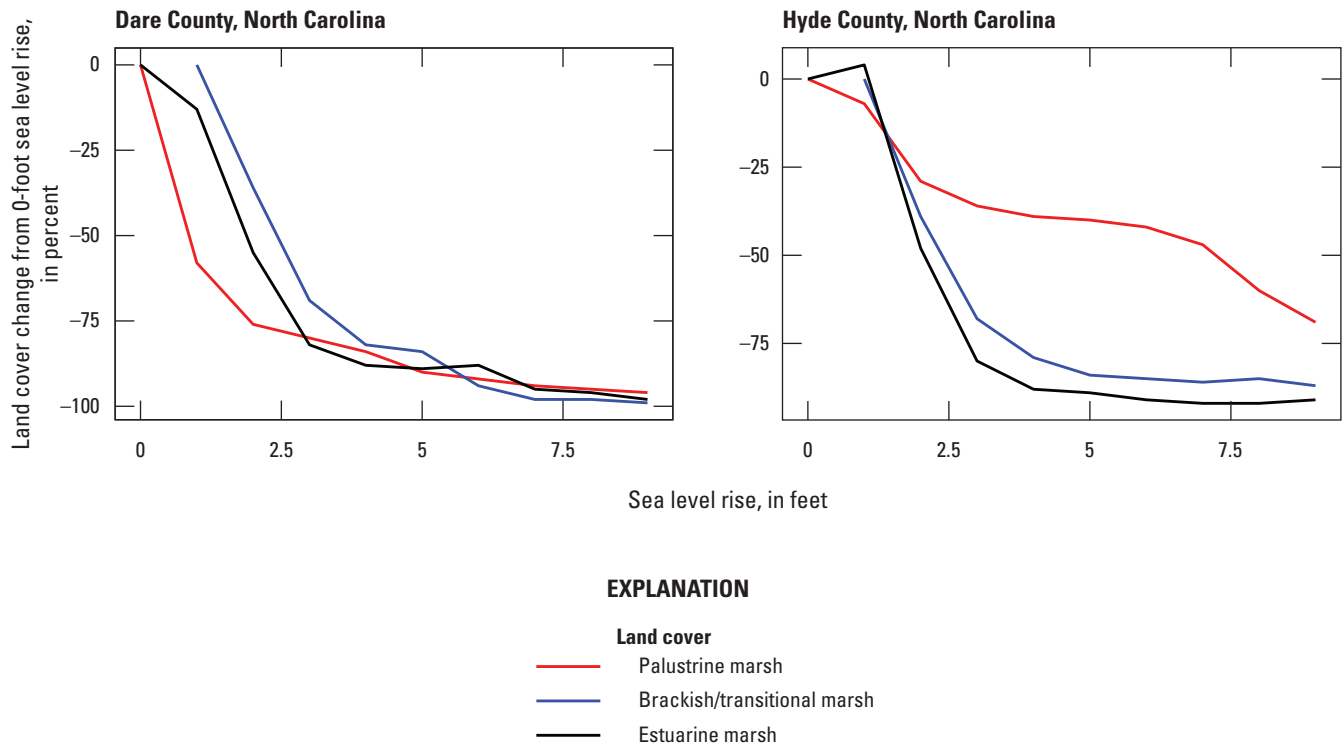


Figure 2.2. Graphs showing percentage change in area of palustrine marsh, brackish/transitional marsh, and estuarine marsh from 0 to 9 feet (ft) of sea level rise above Mean Higher High Water in analysis counties in the North Carolina Middle 2 region ([app. 1](#), [table 1.9](#)). Note that brackish/transitional marsh is not included in the base Coastal Change Analysis Program dataset and therefore has no area in the 0-ft mapping.

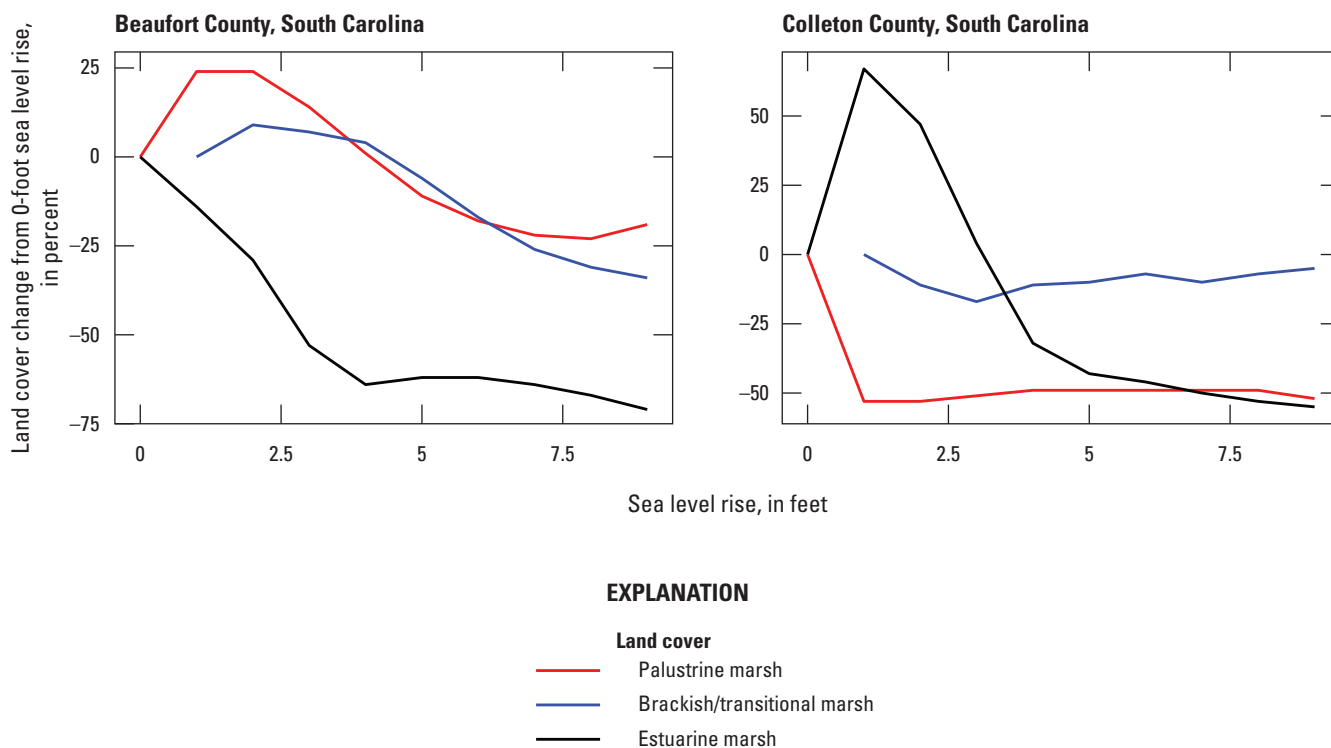


Figure 2.3. Graphs showing percentage change in area of palustrine marsh, brackish/transitional marsh, and estuarine marsh from 0 to 9 feet (ft) of sea level rise above Mean Higher High Water in analysis counties in the South Carolina South region ([app. 1](#), [table 1.9](#)). Note that brackish/transitional marsh is not included in the base Coastal Change Analysis Program dataset and therefore has no area in the 0-ft mapping.

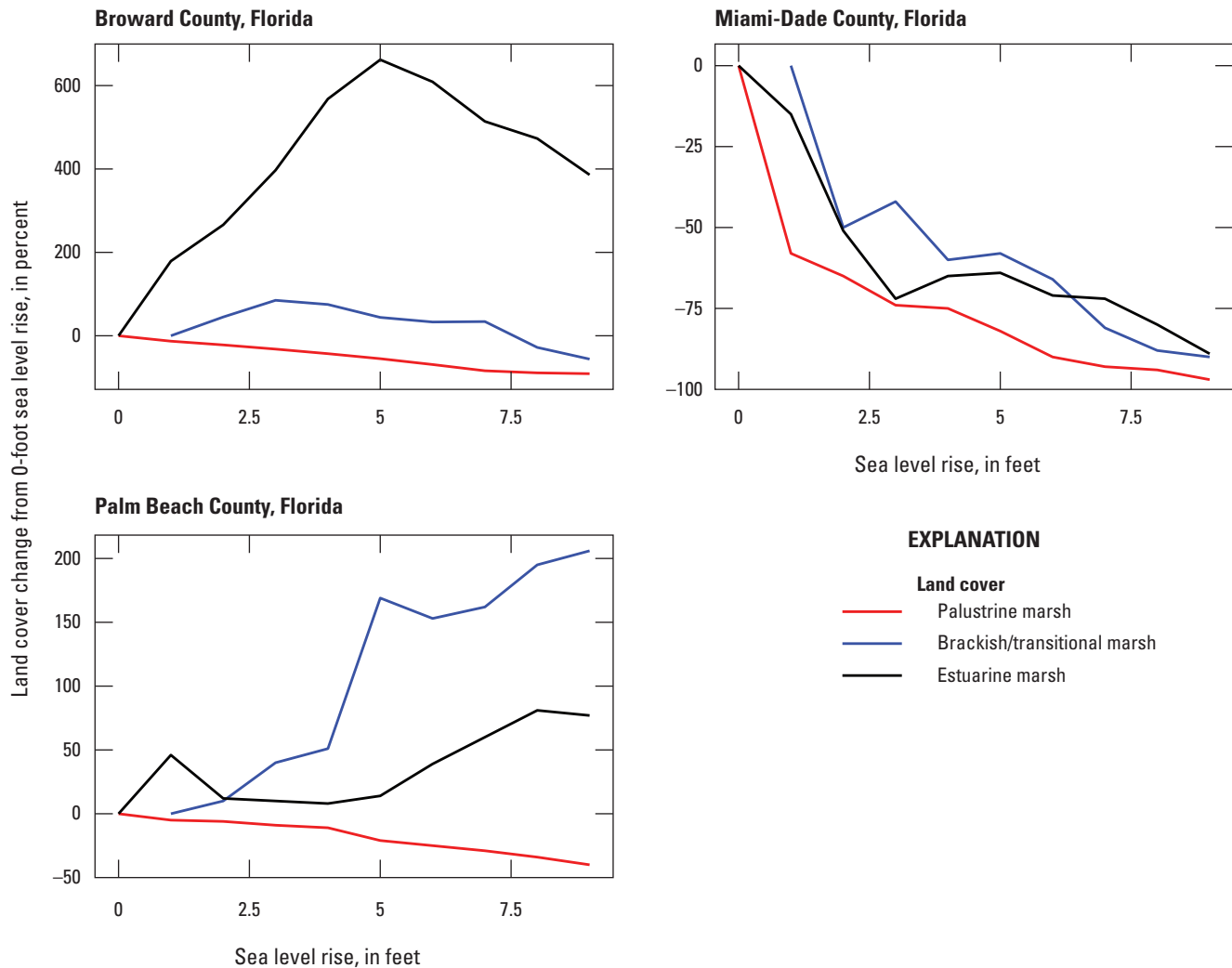


Figure 2.4. Graphs showing percentage change in area of palustrine marsh, brackish/transitional marsh, and estuarine marsh from 0 to 9 feet (ft) of sea level rise above Mean Higher High Water in analysis counties in the Florida Southeast region (app. 1, table 1.9). Note that brackish/transitional marsh is not included in the base Coastal Change Analysis Program dataset and therefore has no area in the 0-ft mapping.

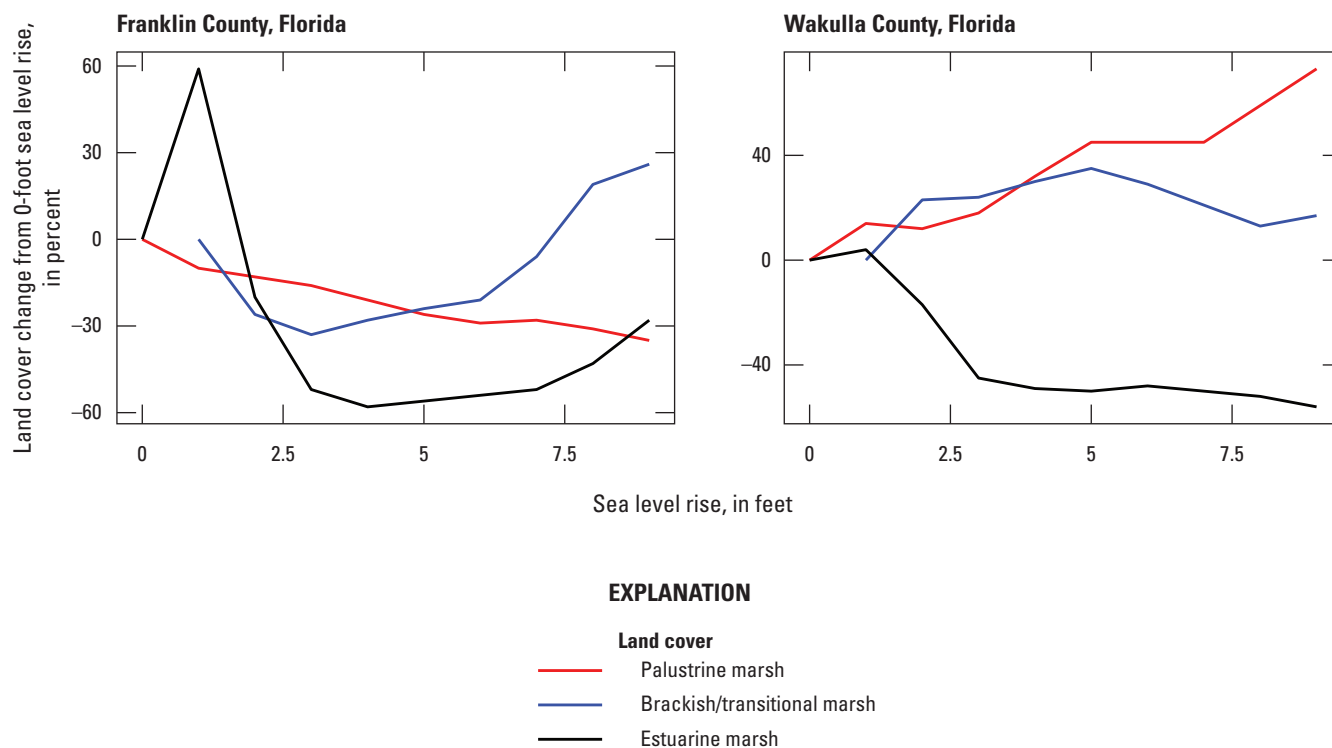


Figure 2.5. Graphs showing percentage change in area of palustrine marsh, brackish/transitional marsh, and estuarine marsh from 0 to 9 feet (ft) of sea level rise above Mean Higher High Water in analysis counties in the Florida Panhandle East region ([app. 1](#), [table 1.9](#)). Note that brackish/transitional marsh is not included in the base Coastal Change Analysis Program dataset and therefore has no area in the 0-ft mapping.

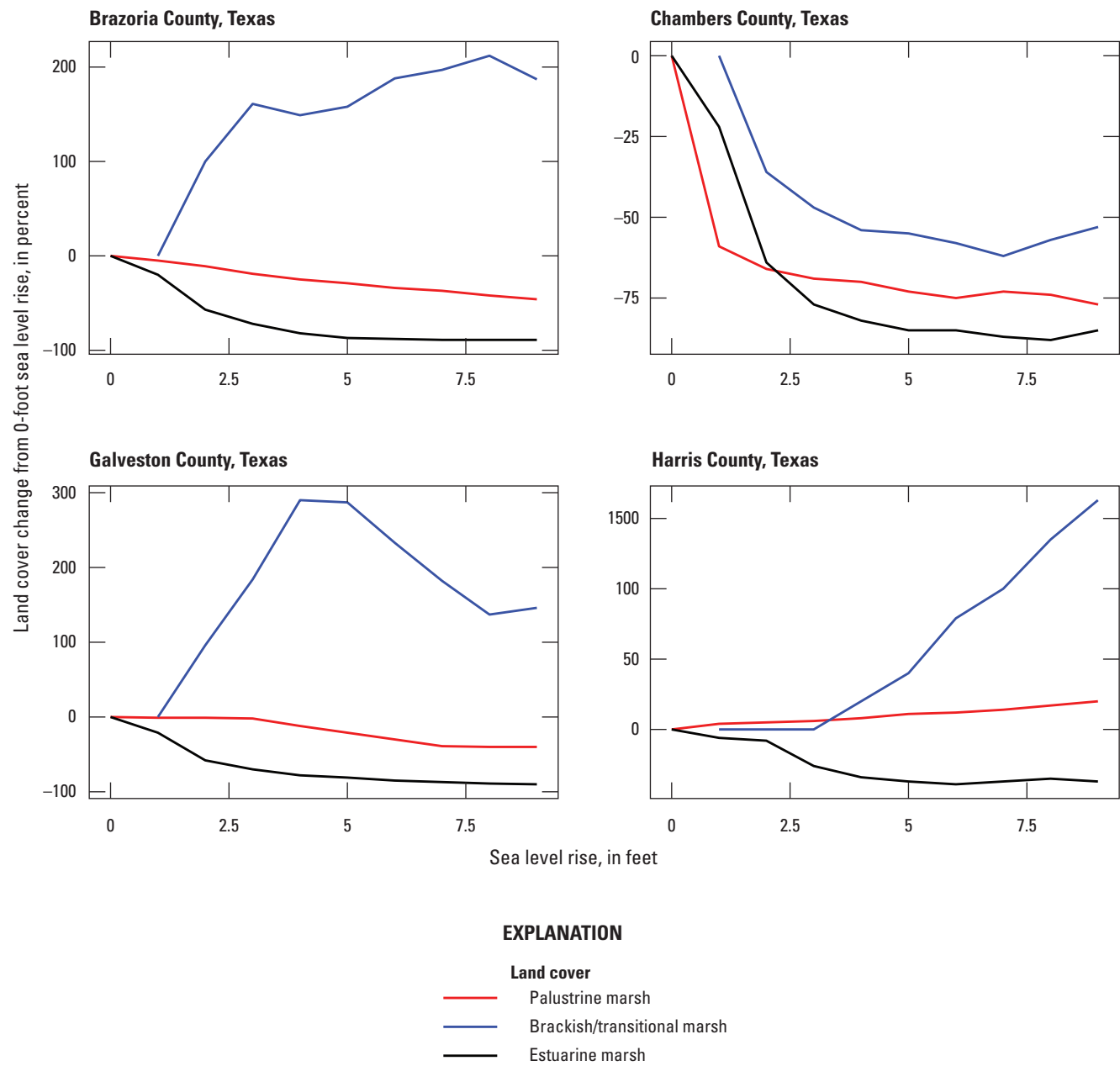


Figure 2.6. Graphs showing percentage change in area of palustrine marsh, brackish/transitional marsh, and estuarine marsh from 0 to 9 feet (ft) of sea level rise above Mean Higher High Water in analysis counties in the Texas North 2 region (app. 1, table 1.9). Note that brackish/transitional marsh is not included in the base Coastal Change Analysis Program dataset and therefore has zero area in the 0-ft mapping.

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